

1.5T HD Head, Neck, and Spine Array and 3.0T Head, Neck, and Spine Array Coils

Operator Manual
2423588



1.5T HD Head, Neck, and Spine Array and
3.0T Head, Neck, and Spine Array Coils
Operator Manual, English
2423588
Rev 4 (09/09)

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Medical Device Directive

This product conforms with the requirements of council directive 93/42/EEC concerning medical devices when they bear the following CE Mark of Conformity:



European Representative:

GE Medical Systems S.C.S.

Quality Assurance Manager

283 rue de la Minière

78530 BUC France

Telephone: +33 1 30 70 40 40



CAUTION:

United States federal law restricts the sale, distribution and use of this device to, or on the order of, a physician.



CAUTION:

This symbol indicates that the waste of electrical and electronic equipment must not be disposed as unsorted municipal waste and must be collected separately. Contact an authorized representative of the manufacturer for information concerning the decommissioning of your equipment.



Coil Description

The 1.5T HD Head, Neck, and Spine Array and the 3.0T Head, Neck, and Spine Array Coils are receive-only 16-channel, 29-element coils designed for use with GE 1.5T MRI Systems and GE 3.0T MRI Systems, respectively. These coils incorporate soft, flexible components that conform to patients' anatomy, accommodating various body contours while minimizing patient discomfort.

Throughout this manual the 1.5T HD Head, Neck, and Spine Array and the 3.0T Head, Neck, and Spine Array coils will be referred to individually as the 1.5T HNS Coil and the 3.0T HNS Coil or collectively as the 1.5T and 3.0T HNS Coils.

The 1.5T and 3.0T HNS Coils are comprised of 4 units: the Head/ Neck Unit (HNU), which includes the anterior (face) unit, the Thoracic/ Lumbar unit (TLU), the Neck Chest Unit (NCU) and Adapter Block (Horseshoe).

The TL Unit, Horseshoe and NCU are identical on systems equipped with HD and P connectors. However, the HNU differs from system to system in the follow ways:

For the 1.5T HNU:

- The HNU with Legacy Bendix connector and HD connector has two 8-channel cables.
- The HNU with one HD connector has one 16-channel cable.
- The HNU with one P connector has one 16-channel cable.

For the 3.0T HNU

- The HNU with 3.0T Legacy connector and HD connector has two 8-channel cables.
- The HNU with one HD connector has one 16-channel cable.
- The HNU with one P connector has one 16-channel cable.

Intended use for the 1.5T HNS Coil

The 1.5T Head, Neck, and Spine Coil is designed for use with 1.5T MRI systems manufactured by GE Healthcare. The coil is for imaging of the brain, cervical spine, soft tissues and vascular of the head, neck, upper chest, thoracic and lumbar spine. The nucleus excited is hydrogen.

Intended use for the 3.0T HNS Coil

The GE 3.0T 16 Channel Head Neck Spine Coil is a receive-only RF coil designed for use with the 3.0T MRI Systems manufactured by GE Healthcare. The coil is intended for imaging of the brain; cervical, thoracic and lumbar spines; as well as the soft tissues and vasculature of the head, neck and upper chest. The nucleus excited is hydrogen.

Contraindications

Patients with pacemakers and other implanted devices should not be allowed access into the MR scanner. Individual safety screening must be conducted by MR trained and qualified personnel.

Please refer to the MR Safety Guide, Part #2381696 for additional and comprehensive MR safety information

Manual Notes

This operator manual does not apply to the 1.5T HD Head, Neck and Spine, Part #2416329, revision 2.

This operator manual does not apply to the 3.0T Head, Neck and Spine that is manufactured by Medrad and is made up of part numbers #5325953-2 (Head Neck Unit), 5325953-3 (Thoracic Lumbar Unit) and 5325953-4 (Neck Chest Unit).

The user must read and understand all instructions in this manual as well as the MR Safety Guide, part #2381696, before attempting to use the product. Please request training from GE if needed.

This manual should be kept with the equipment and be readily available at all times. It is important for you to periodically review the procedures and safety precautions. **It is important for you to read and understand the contents of this manual before attempting to use this product.**

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Chapter 1

Introduction and Shipping Contents

Introduction

This manual describes the safety precautions, features, use and care of the 1.5T HNS Coil compatible with 1.5T GE MRI Systems with P connectors, HD connectors and Legacy Bendix connector, and the 3.0T HNS Coil compatible with 3.0T GE MRI Systems with P connectors, HD connectors and 3.0T Legacy connector. Please review this manual thoroughly before using the device for imaging. Refer to your MR scanner manual in conjunction with this manual.

If you have questions or comments on this manual, or need assistance with the use of this product, please contact your GE Healthcare representative.

Use of Symbols

Coil users should understand the following symbols used for caution and warning explanations:



CAUTION:

Indicates a hazard that may cause a minor injury or property damage if instructions are not followed.



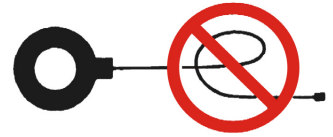
WARNING:

Indicates a hazard that may cause severe personal injury, death, or substantial property damage if instructions are not followed.



WARNING:

Do not cross or loop coil cables. Looping the cable can result in arcing which may damage the coil and/or cause patient burns. Do not form loops with RF receive coil cables and ECG leads.



CAUTION:

This symbol indicates that the waste of electrical and electronic equipment must not be disposed as unsorted municipal waste and must be collected separately. Contact an authorized representative of the manufacturer for information concerning the decommissioning of your equipment.



Symbol and Label Definition

Table 1-1 Symbol and Label Definition

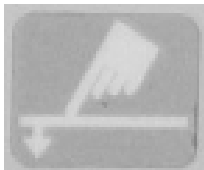
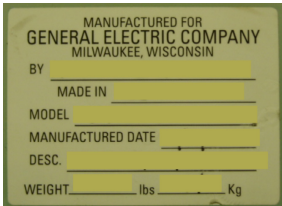
Symbol	Definition
	Manufacturer
	Consult Accompanying Documents
SN	Serial Number
REF	Part Number
	This label, affixed to the HNU Anterior and the Horseshoe, informs the user to firmly press down on all coil components to ensure that they are fully engaged and connected. Failure to do so may cause artifacts and decrease image quality.
	Do not insert connector with this side up. If connector is inserted with this side up, the coil will not operate.
	Phantom Identifier

Table 1-1 Symbol and Label Definition (Continued)

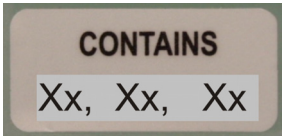
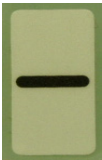
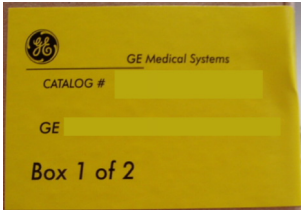
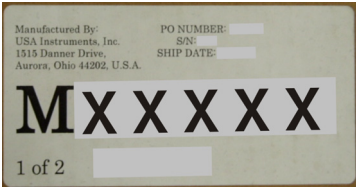
Symbol	Definition
	Phantom Contents
	Phantom Positioning
	Packaging CE label
	Do not allow package to freeze
	Do not drop or handle package roughly

Table 1-1 Symbol and Label Definition (Continued)

Symbol	Definition
	Keep packaging dry. Do not drop or handle package roughly. The arrow points to the side of the package that is to remain facing up.
	Transportation and Storage Specifications
	Shipping Content label. Identifies the catalog number, GE part number and the number of boxes in the shipment.
	Identifies contents, purchase order number, part number, quantity, shipping slip number, serial number and date shipped.
	Shipping label. Identifies manufacturer, purchase order number, serial number, ship date and part numbers and number of boxes in the shipment.

Coil Classification

Symbol	Classification
	Class II equipment
	Type BF applied

- IPX 0, not rated against water ingress protection
- Suitable for continuous operation

Shipping Contents: 1.5T HNS - Legacy Bendix and HD Connector Coil (3 Connectors)

Box 1 (HNS Coil #2418627)

Part Name	Part Number	Qty.
HD Head, Neck and Spine Array	2416329	1
HNS Sys Cable Excite III, 8ch	2423004	1
HNU Sys Cable Excite II, 8ch	2423006	1
Assembly Mirror	2423315	1
Pad Kit	2418081	1
Operator Manual Packet	2423588-500	1

Box 2

Part Name	Part Number	Qty.
Phantom Kit - head section and positioner	2418236 (2420112 - Positioner, 2424360 - Head Phantom, 2422888 - Neck Phantom)	1

Box 3 and 4

Part Name	Part Number	Qty.
Phantom Kit - TL component	2424359 (TL phantom: 2424361)	1

Shipping Contents: 1.5T HNS - HD Connector Coil (2 Connectors)

Box 1 (HNS Coil #2418626)

Part Name	Part Number	Qty.
HD Head, Neck and Spine Array	2416329	1
HNU Sys Cable Excite III, 16ch	2423003	1
Assembly Mirror	2423315	1
Cable Exit Plug	2423029	1
Pad Kit	2418081	1
Operator Manual Packet	2423588-500	1

Box 2

Part Name	Part Number	Qty.
Phantom Kit - Head Section and Positioner	2418236 (2420112 - Positioner, 2424360 - Head Phantom, 2422888 - Neck Phantom)	1

Box 3 and 4

Part Name	Part Number	Qty
Phantom Kit - TL component	2424359 (TL phantom: 2424361)	1

Shipping Contents: 1.5T HNS - HD and P Connector Coil (2 Connectors)**Box 1 (HNS Coil #2424381)**

Part Name	Part Number	Qty.
HD Head, Neck and Spine Array	2416329	1
HNU Sys Cable P connector, 16ch	2423927	1
Assembly Mirror	2423315	1
Pad Kit	2418081	1
Operator Manual Packet	2423588-500	1

Box 2

Part Name	Part Number	Qty.
Phantom Kit - Head Section and Positioner	2418236 (2420112 - Positioner, 2424360 - Head Phantom, 2422888 - Neck Phantom)	1

Box 3 and 4

Part Name	Part Number	Qty
Phantom Kit - TL component	2424359 (TL phantom: 2424361)	1

Shipping Contents: 3.0T HNS Coil - 3.0T Legacy and HD Connector Coil (3 Connectors)

Box 1 (HNS Coil #M3335HD)

Part Name	Part Number	Qty.
3.0T 16 Channel HNS Coil	5341333	1
Assy, 8CH EXC III HNU SYS CBL, 3.0T	5338551	1
System Cable, 8CH EXCITE II, HNU, 3.0T HNS	5344575	1
Assembly Mirror	2423315	1
Pad Kit	2418081	1
Operator Manual Packet	2423588-500	1
Coil Definitions Disk	5325975	1
3.0T HNS Service Manual	5367722	1

Box 2

Part Name	Part Number	Qty.
Phantom Kit - Head Section and Positioner	5342611 (2420112 - positioner, 2413435 - head phantom, 2413436 - neck phantom)	1

Box 3 and 4

Part Name	Part Number	Qty.
Phantom Kit - TL Component	5342613 (TL phantom - 2381682-10)	1

Shipping Contents: 3.0T HNS Coil - HD Connector Coil (2 Connectors)

Box 1 (HNS Coil #M3335HC)

Part Name	Part Number	Qty.
3.0T 16 Channel HNS Coil	5341333	1
16CH EX III HNU SYS CBL, 3.0T	5338550	1
Assembly Mirror	2423315	1
Cable Exit Plug	2423029	1
Pad Kit	2418081	1
Operator Manual Packet	2423588-500	1
Coil Definitions Disk	5325975	1
3.0T HNS Service Manual	5367722	1

Box 2

Part Name	Part Number	Qty.
Phantom Kit - Head Section and Positioner	5342611 (2420112 - positioner, 2413435 - head phantom, 2413436 - neck phantom)	1

Box 3 and 4

Part Name	Part Number	Qty.
Phantom Kit - TL Component	5342613 (TL phantom - 2381682-10)	1

Shipping Contents: 3.0T HNS Coil - HD and P Connector Coil (2 Connectors)**Box 1 (HNS Coil #M7000AP)**

Part Name	Part Number	Qty.
3.0T 16 Channel HNS Coil	5341333	1
System Cable, 16CH 3.0T HNS, P Connector	5338113	1
Assembly Mirror	2423315	1
Cable Exit Plug	2423029	1
Pad Kit	2418081	1
Operator Manual Packet	2423588-500	1
Coil Definitions Disk	5325975	1
3.0T HNS Service Manual	5367722	1

Box 2

Part Name	Part Number	Qty.
Phantom Kit - Head Section and Positioner	5342611 (2420112 - positioner, 2413435 - head phantom, 2413436 - neck phantom)	1

Box 3 and 4

Part Name	Part Number	Qty.
Phantom Kit - TL Component	5342613 (TL phantom - 2381682-10)	1

Coil installation and service

Coil installation and configuration must be done by your GE Service Representative. GE Field Engineers shall consult the appropriate Service Method CD for installation instructions.

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Chapter 2

Safety

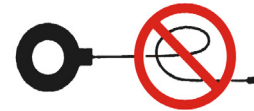
Patient Safety

Patient safety and comfort must be your primary concern during the scanning procedure. Always follow proper safety, operating and maintenance procedures to ensure that the patient is not exposed to electrical or mechanical hazards with the potential to cause injuries.



WARNING:

Do not cross or loop coil cables. Looping the cable can result in arcing which may damage the coil and/or cause patient burns. Do not form loops with RF receive coil cables and ECG leads.



Route cables through the center of the magnet bore. Place cables under the cushion whenever possible. Routing near the sides of the bore increases the likelihood of cable heating from induced currents.

Keep the length of the cable in the bore to a minimum. Avoid bending the cable 180 degrees. Route cables out of the bore in the most direct way possible, without looping or coiling.

Use a thermal resistant material or pad to keep the cable from touching the patients.



WARNING:

Do not allow the coil cable to come into contact with the patient's skin. Use a thermal resistant material or pad to keep the cable from touching the patient. Failure to do so may result in patient burns.



WARNING:

RF can cause localized heating at contact points between adjacent body parts when a loop is formed. Such localized heating can result in a tingling sensation, discomfort, or burns. This could occur when a patient's hands are touching or when a female patient's breasts are compressed to her chest. Use pads between body parts to avoid creating a loop with adjacent body parts.



WARNING:

RF can cause localized heating at contact points between the patient and the bore resulting in discomfort or burns. Place appropriate nonconductive padding between the patient and the bore wherever a portion of the body may come into contact with the magnet opening.



WARNING:

This coil is not suitable for use in the presence of a flammable anesthetic mixture with air or with oxygen or nitrous oxide.



WARNING:

Always place appropriate non-conductive padding between the surface coil and the skin of the patient to prevent burn injuries.



WARNING:

Prior to patient placement in the coil, assure that any breached or compromised patient skin surfaces that come in contact with the coil have been adequately bandaged or covered.

**CAUTION:**

To avoid injuring the patient or damaging the coil, watch carefully for pinch points as the table moves into the bore. Stop advancing the table if the patient or any part of the coil comes into contact with the bore.

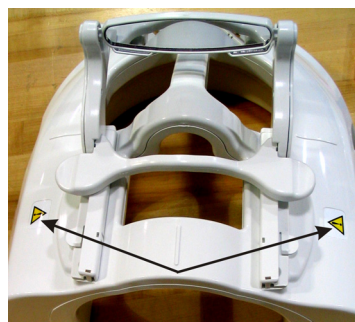
Ensure that the patient is comfortably positioned.

Monitor the patient periodically. If the patient reports heating, burning or tingling sensations, **STOP THE SCAN IMMEDIATELY.**

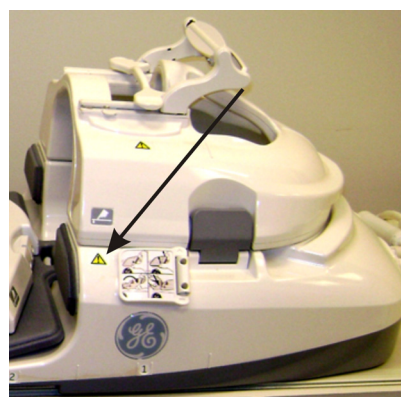
Warning and Caution Label locations on the coil

**WARNING:**

The arrows on the photograph to the right point towards two warning labels located on the HNU anterior unit. Do not allow the patient to place their hands on the anterior unit of the coil when the table is moving into the bore. The hands or arms of the patient may be pinched between the coil and the bore.

**WARNING:**

The arrow on the photograph to the right points toward a warning label located on the HNU posterior unit. When scanning patients *without* the chest component, always place the connector cover securely over the connector on the Posterior Head Component. The cover prevents the patient's hand from coming in contact with the connector. Failure to comply will leave the connector exposed, placing the patient at risk for RF burns or shock if the patient touches the connector during a scan.



Equipment Safety

All personnel using this coil must be instructed in its proper connection and handling. Personnel must observe all warnings and cautions appearing in this Manual.



WARNING:

Auxiliary equipment (e.g. gating equipment, vital signs monitoring systems and RF coils) that has not been specifically tested and approved for use in the MR environment may result in burns or other injuries to the patient. Such equipment may also interfere with proper operation of this coil.



CAUTION:

Looping may cause RF coupling and degrade the coil's scan performance



WARNING:

Do not use accessories (e.g. pads or straps) that have not been specifically tested and approved for use in the MR environment. Use of non-approved accessories may result in patient burns or injuries or image degradation.

Electrical and Mechanical Safety



WARNING:

Chemical exposure risk. This product contains chemicals, including lead, known to the State of California to cause birth defects or other reproductive harm. Wash hands after handling coil.



WARNING:

This coil consists of electrical and mechanical components. Service personnel must have specialized training to ensure the safe operating condition of the coil. Only properly trained and qualified personnel should service the coil.

The electrical and mechanical assemblies and parts of the coil must be used with care and should be routinely inspected.



WARNING:

Before using the coil, visually check it to ensure that there is no external damage. Do not use the coil if the housing or cable is broken.

Modifications

Any changes or modifications to this coil must be approved and performed by GE Healthcare prior to installation.

Reporting Incidents

Users should immediately report to GE Healthcare any incidents or injuries that result from coil operation involving a patient, operator or maintenance employee.

If an accident occurs as a result of coil operation, do not operate the equipment until an authorized investigation has been conducted.



CAUTION:

Users should place a service call any time the coil is dropped or mishandled. A GE Service Representative should inspect the coil after it has been dropped to ensure it is safe to use.

Emergency Procedures

In the unlikely event that a coil creates smoke, sparks or makes an unusually loud noise, or if the patient requires emergency assistance:

- Stop the scan if one is in progress
- Remove the patient from the scan room if medical treatment is needed.

NOTE: It is recommended to unplug the system connector prior to removing the patient.

- Discontinue use of the coil until a GE Service Representative can confirm that it is safe to operate.

MR Safety Information



CAUTION:

This product contains chemicals, including lead, known to the state of California to cause birth defects or other reproductive harm. Wash hands after handling.

Please refer to the MR Safety Guide, part #2381696. The MR Safety Guide describes the safety information you and the physicians must understand thoroughly before you begin to use the MR system. If you need additional training, seek assistance from qualified GE Healthcare personnel.

This equipment is intended for use by qualified personnel only.

This manual should be kept with the equipment and be readily available at all times. It is important for you to periodically review the procedures and safety precautions. **It is important for you to read and understand the contents of this manual before attempting to use this product.**

Chapter 3

Using the HNS Coils

The 1.5T and 3.0T HNS Coils enable the clinical user to collect high resolution images of the head, neck, and spine.

NOTE: The HNS Coil is available for the software release version 14.0 and greater for the 1.5T coil and 15.x and greater for the 3.0T coil. It is important to make certain that your system coil connector is compatible with the connector on your coil. It is also important that you use the mode configuration names appropriate for the software you are running, otherwise faults may occur.

Coil Specifications

Field strength	Model	Device #	Frequency
1.5T	HD Head, Neck and Spine Array	2416329	63.86 MHz
3.0T	Head, Neck, and Spine Array	5341333	127.73 MHz



WARNING:

Do not allow the coil cables to touch the patient. Use a thermal resistant material or pad to keep the cable from touching the patient. Failure to comply may cause patient burns.



CAUTION:

Unplug the coil before lowering the table. Failure to do so may cause damage to the coil and connector and may render the coil inoperable.



CAUTION:

Do not carry the coil by the cables. Damage to the coil may occur, rendering the coil inoperable.



CAUTION:

Do not pick up or carry the Head Component by the mirror attachment. To avoid damaging the coil, pick up and carry the Head Component using two hands on the bottom of the coil.

Installing the mirror onto the HNU Anterior (Face) Unit

To prevent breakage, the mirror is designed to break away from the anterior unit. In the event that the mirror assembly is pulled from the anterior unit, the mirror can be reinstalled using the following procedure:

NOTE: If the mirror assembly has come apart or the snap features on the top rails are broken, contact your GE service representative for replacement. A top mirror that gets pulled out of a socket is an example of a failure that may occur. See figure 3-1 for an example. If this occurs, please contact your GE service representative for replacement or repair.

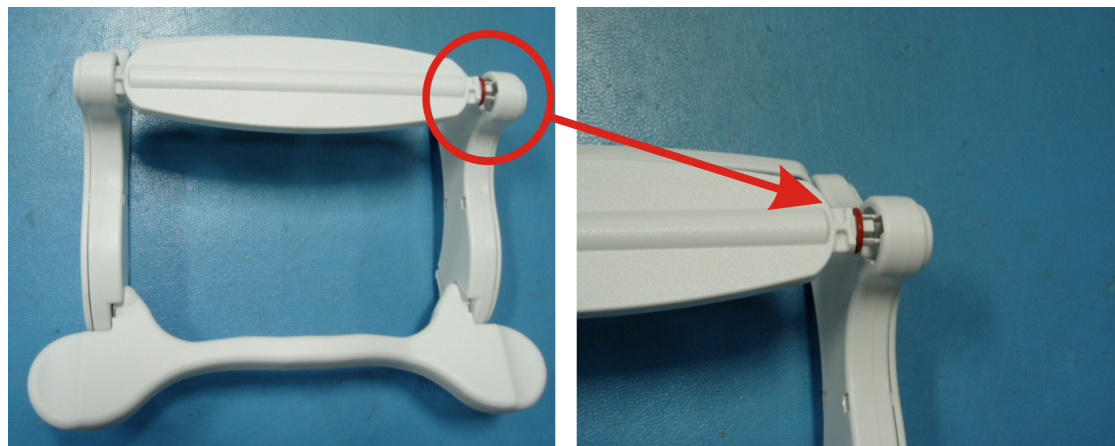


Figure 3-1 Example of the top mirror pulled out from the socket

1. If the mirror has come off of the anterior unit and the top rails (shown in figure 3-1) are still attached to the mirror, push the locking bar on the mirror into the up position (shown in figure 3-9). This will release the top rails from the mirror so that the mirror can be re-installed onto the anterior unit as described in the following steps.
2. Attach the top rails (shown in figure 3-2) onto the bottom rails that are screwed onto the anterior unit (figure 3-3). The arrows in figure 3-3 point towards the rails.

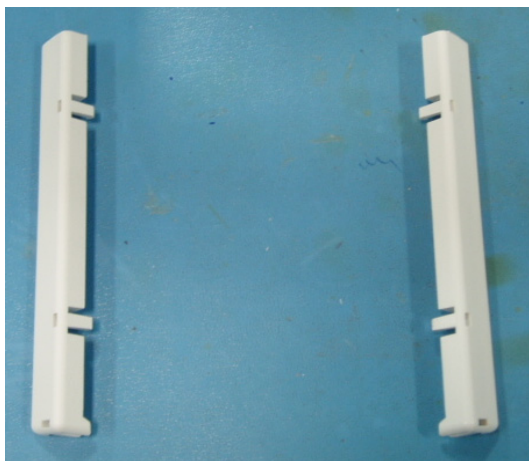


Figure 3-2 Top Rails



Figure 3-3 Arrows point toward the bottom rails that are attached to the anterior unit

NOTE: Steps 3 and 4 instruct how to install the right top rail onto the right bottom rail. To attach the left top rail, simply follow the same procedure, but use the pieces for the left side. The part numbers for the top rails can be helpful to distinguish the two top rails apart. The part number for the right top rail is 2419526, and the part number for the left top rail is 2419525. Figure 3-4 shows the location of the part numbers on the rails.

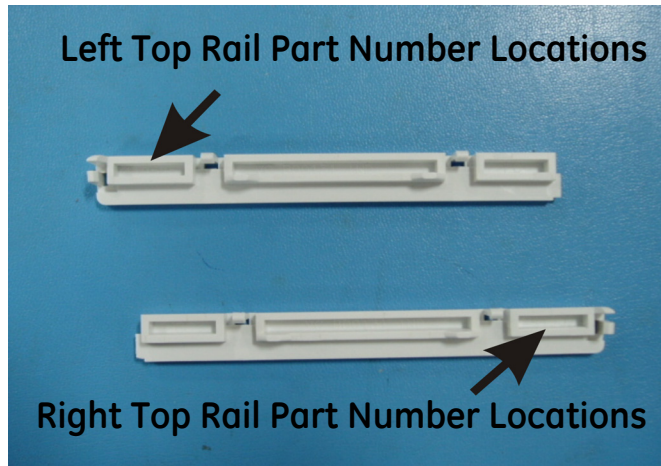


Figure 3-4 Top rail part number locations

3. Insert the front tab on the top right rail into the front cut-out of the bottom rail (see figure 3-5 and 3-6).

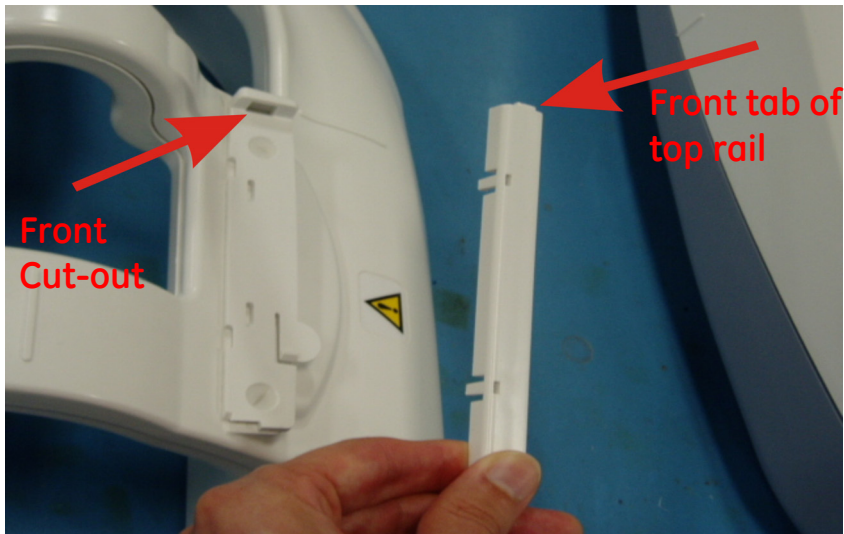


Figure 3-5 The front tab of the top rail and the front cut-out of the bottom rail



Figure 3-6 Inserting the top rail tab into the bottom rail

4. Gently push the top rail down onto the bottom rail (figure 3-7). A click will be heard when the top rail is secured to the bottom rail. Figure 3-8 shows the top rail on the bottom rail.

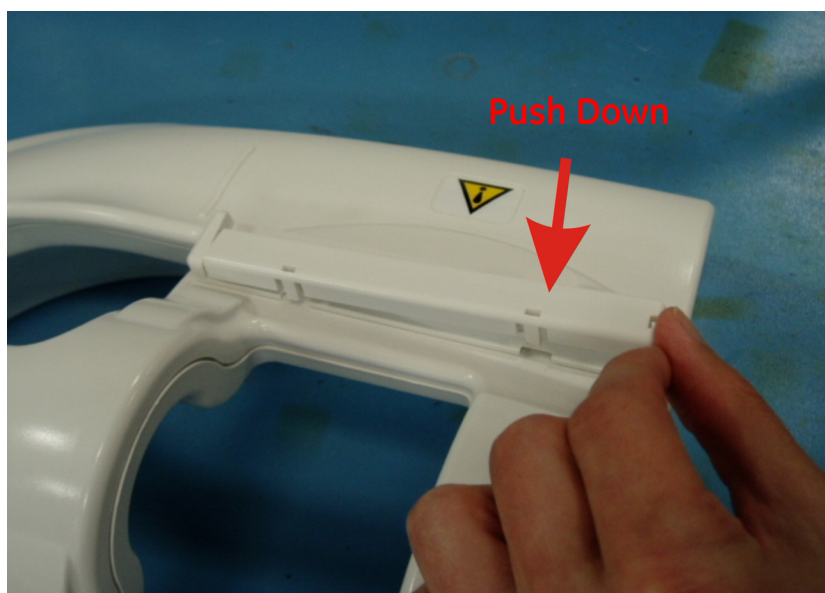


Figure 3-7 Gently push the top rail onto the bottom rail



Figure 3-8 Top rail on the bottom rail

5. If needed, attach the opposite rail using the same method as described in steps 3 and 4.

6. Push the locking bar of the mirror into the up position. See figure 3-9.

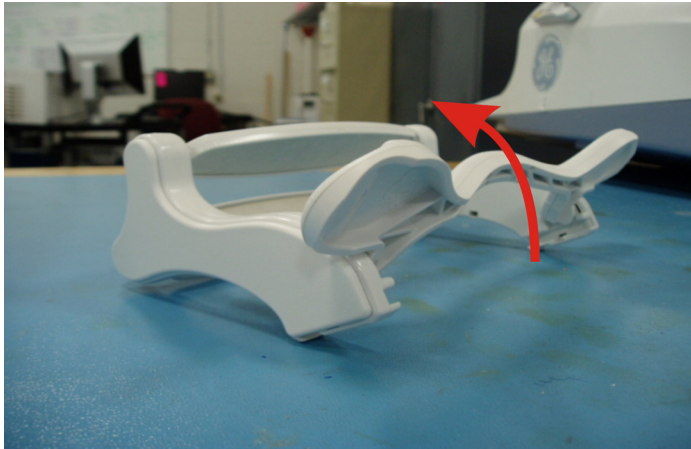


Figure 3-9 Locking bar in the up position

7. Align the channels at the bottom of the left and right side of the mirror with the channel of the left and right rails. See figure 3-10.



Figure 3-10 Mirror channel aligned with the rail channels

8. Slide the mirror onto the rails as shown in figure 3-11. The mirror needs to slide past the rail tabs. See figure 3-12 for a photograph of a rail tab.

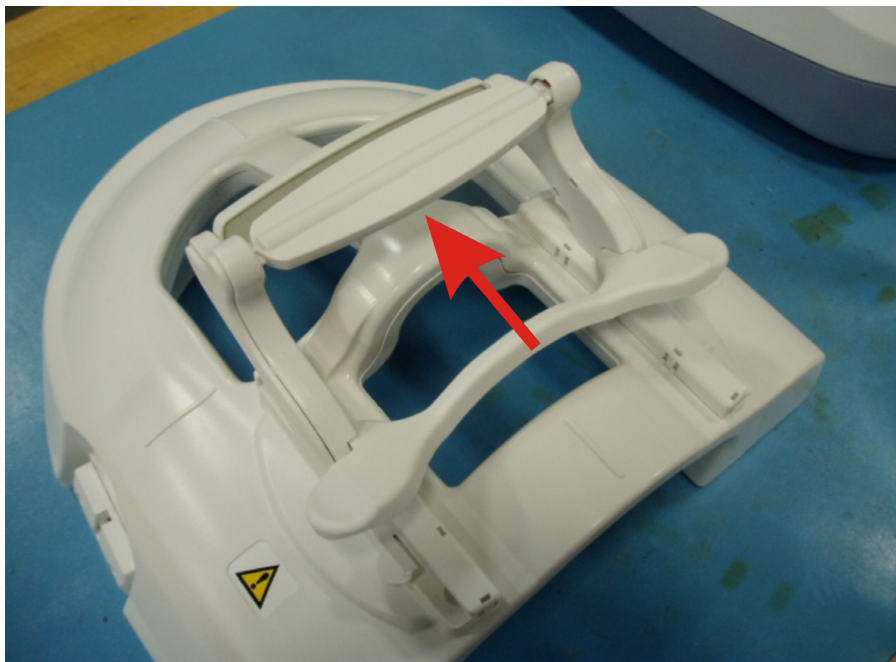


Figure 3-11 Slide mirror onto the rails

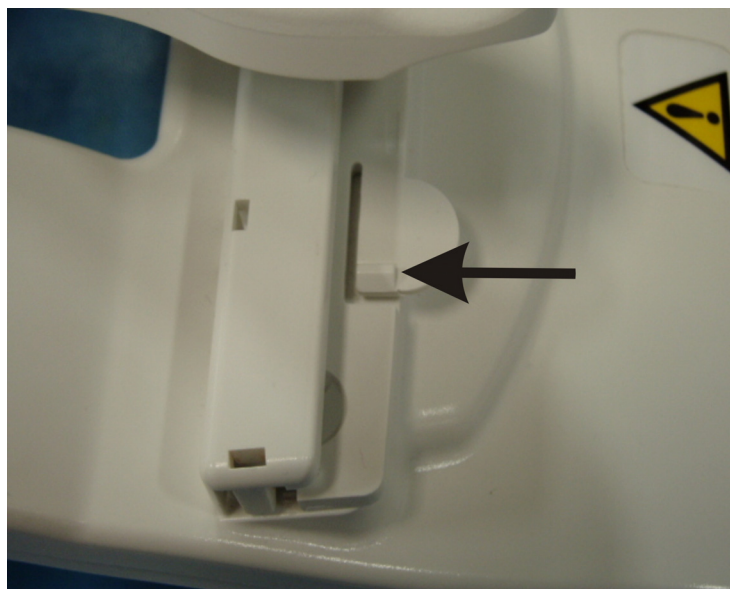


Figure 3-12 Right side rail tab.

9. Gently push the locking bar on the mirror down to lock the mirror in place. See figure 3-13.

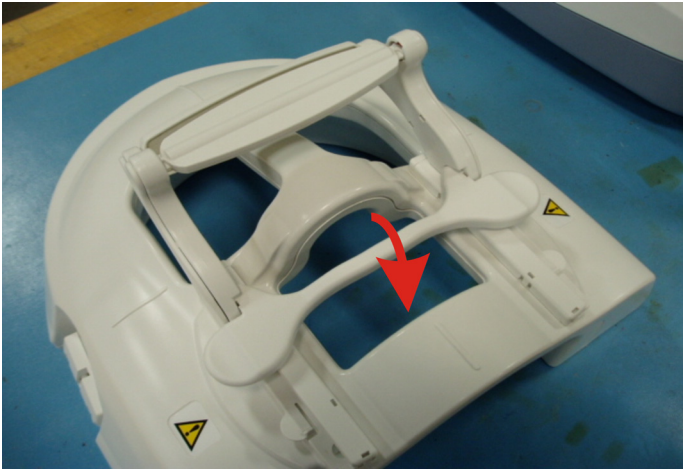


Figure 3-13 Push the locking bar down

10. The mirror is now installed onto the anterior unit.

Coil Components

The 1.5T and 3.0T HNS Coils are composed of five separate components. The element groups are marked 1-6 on the sides of the coil for positioning and landmarking purposes (please refer to Figure 3-14 below):

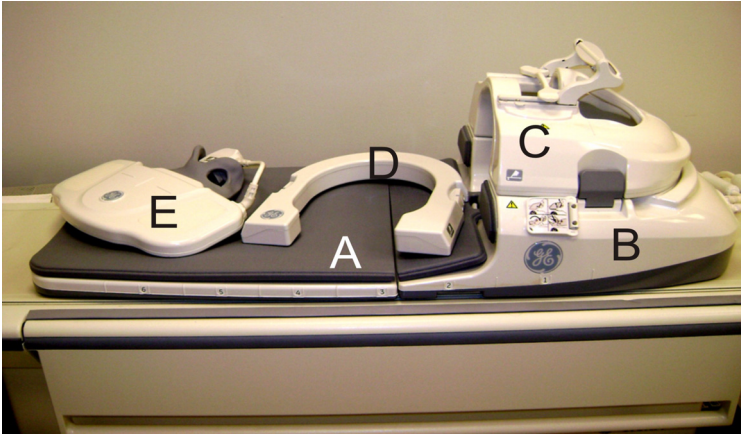


Figure 3-14 Components of the HNS (the groups refer to the element distribution of the coil. See Figure 3-15 for element distribution):

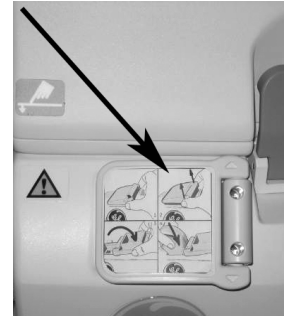
- A = TL Spine (TL) (groups 3 through 6)
- B = HNU Posterior Unit (Posterior Head and groups 1 and 2)
- C = HNU Anterior Unit (Face)
- D = Anterior Adapter Block (Horseshoe)
- E = Neck Chest Unit (Chest)

The components may be used all together, or in several combinations. The TL component may be used independently. Please refer to Table 3-1 for more information on components and configurations.



WARNING:

When scanning patients *without* the chest component, always place the connector cover securely over the connector on the Posterior Head Component. The cover prevents the patient's hand from coming in contact with the connector. Failure to comply will leave the connector exposed, placing the patient at risk for RF burns or shock if the patient touches the connector during a scan.



CAUTION:

The Revision 3 1.5T HNS Coil is not compatible with HNS Coil Revision 2 components, with the exception of the TL Spine, which is interchangeable. Connection between incompatible parts may be physically possible, but faults will occur. Check the label on each component for the revision number.



CAUTION:

1.5T HNS and 3.0T HNS coil components are not compatible. Check the labels on each component to ensure common field strength. Connection between incompatible parts may be physically possible, but faults will occur.



CAUTION:

The 3.0T HNS coil components manufactured by GE Healthcare Coils and the 3.0T HNS coil components manufactured by Medrad are not compatible. Check the labels on each component for common manufacturer's name before use. Using 3.0T HNS Medrad components with 3.0T HNS GE Healthcare Coil components may degrade performance or safety.

Element Distribution

There are a total of 29 elements in the 1.5T and 3.0T HNS Coils. Fig. 3-15 depicts the locations of the groups of elements. Use this information along with Table 3-1 on the following page to determine the correct configuration for your study.

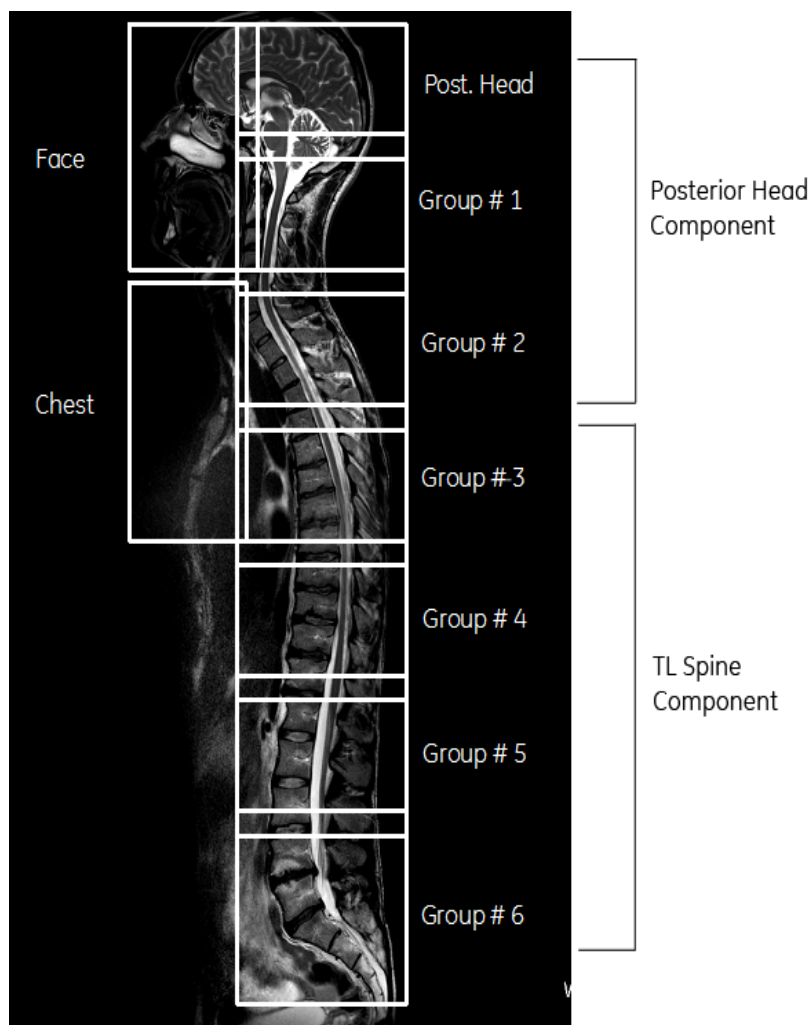


Figure 3-15 1.5T and 3.0T HNS Coil element distribution

Coil Mode Configuration

The 1.5T and 3.0T HNS Coils were designed to eliminate multiple coil usages per patient in order to increase throughput and patient comfort. These coils have 39 different configurations with which to scan. These configurations allow you to optimize field of view (FoV) when using a variety of coil components, and will enable you to scan the region of interest with excellent image quality without mandatory coil changes and patient disruption.

To optimize image quality, use the following section as a guide to help you select the proper components and coil configurations for your study.

The following Configuration Table explains the different coil components and configurations available to you. When the desired coil components are plugged into the system, the available configurations become selectable. Therefore, hardware (coil components) dictates the coil configurations that are available.

Select a configuration based on the anatomy of interest, and the FoV desired. Then center the anatomy to the element groups used in the desired configuration.

Table 3-1 Coil Configurations

Components Connected	Configurations Available	Max. FoV	Best Use	Element Groups Used	Asset
Posterior Head, Horseshoe and TL Spine	HNS CTLTOP	48cm	Largest FoV C and T spine	1,2,3 and 4	No
	HNS CTLMID	48cm	Largest FoV T and L spine	2,3,4 and 5	No
	HNS CTLBOT	48cm	Largest FoV T and L spine	3,4,5 and 6	No
	HNS CTL123	40cm	Large FoV C and T spine	1,2 and 3	No
	HNS CTL234	40cm	Large FoV T spine	2,3 and 4	No
	HNS CTL345	40cm	Large FoV T and L spine	3,4 and 5	No
	HNS CTL456	40cm	Large FoV T and L spine	4,5 and 6	No
	HNS CTL12	28cm	Smallest FoV C spine	1 and 2	No
	HNS CTL23	28cm	Smallest FoV Upper T spine	2 and 3	No
	HNS CTL 34	28cm	Smallest FoV Mid/Low T spine	3 and 4	No
	HNS CTL45	28cm	Smallest FoV Lower T/L spine	4 and 5	No
	HNS CTL56	28cm	Smallest FoV L spine	5 and 6	No
Posterior Head and Horseshoe	HNS CTL12	28cm	Smallest FoV C spine	1 and 2	No

Table 3-1 Coil Configurations (Continued)

Components Connected	Configurations Available	Max. FoV	Best Use	Element Groups Used	Asset
Posterior Head, Face and TL spine	HNS HEAD	30cm	Brain	Post Head, Face and 1	Yes
	HNS 35FOV	48cm	Brain, C spine	Post Head, Face and 1,2 and 3	No
	HNS 48FOV	48cm	Brain, C spine, T spine	Post Head, Face and 1,2,3 and 4	No
	HNSp CTLTOP	48cm	Largest FoV C and T spine	1,2,3 and 4	No
	HNSp CTLMID	48cm	Largest FoV T and L spine	2,3,4 and 5	No
	HNS CTLBOT	48cm	Largest FoV T and L spine	3,4,5 and 6	No
	HNSp CTL123	40cm	Large FoV C and T spine	1,2 and 3	No
	HNSp CTL234	40cm	Large FoV T spine	2,3 and 4	No
	HNS CTL345	40cm	Large FoV T and L spine	3,4 and 5	No
	HNS CTL456	40cm	Large FoV T and L spine	4,5 and 6	No
	HNSp CTL12	28cm	Smallest FoV C spine	1 and 2	No
	HNSp CTL23	28cm	Smallest FoV Upper T spine	2 and 3	No
	HNS CTL34	28cm	Smallest FoV Mid/Low T spine	3 and 4	No
	HNS CTL45	28cm	Smallest FoV Lower T/L spine	4 and 5	No
	HNS CTL56	28cm	Smallest FoV L spine	5 and 6	No
Posterior Head and Face	HNSp CTL12	28cm	Smallest FoV C spine	1 and 2	No
	HNS HEAD	30cm	Brain	Post Head, Face and 1	Yes

Table 3-1 Coil Configurations (Continued)

Components Connected	Configurations Available	Max. FoV	Best Use	Element Groups Used	Asset
Posterior Head, Face and Chest	HNS Neck	30cm	Smallest FoV Carotids, C spine, ST Neck	Face, 1,2 and 2 Chest elements	Yes
	HNS NeckChest	35cm	Large FoV Carotids, C spine, ST Neck	Face, 1, 2 and all 5 Chest elements	Yes
	HNS NVFull	48cm	Largest FoV Carotids, C spine, ST Neck	Post Head, Face, 1,2 and all 5 Chest Elements	Yes
	HNS NVHead	30cm	Brain	Post Head, Face and 1	Yes
	HNS-CTL12	28cm	Smallest FoV C spine	1 and 2	No

Table 3-1 Coil Configurations (Continued)

Components Connected	Configurations Available	Max. FoV	Best Use	Element Groups Used	Asset
Posterior Head, Face, TL Spine and Chest	HNS Neck	30cm	Smallest FoV Carotids, C spine, ST Neck	Face, 1,2 and 2 Chest elements	Yes
	HNS NeckChest	35cm	Large FoV Carotids, C spine, ST Neck	Face, 1,2 and all 5 Chest elements	Yes
	HNS NVFull	48cm	Largest FoV Carotids, C spine, ST Neck	Post Head, Face, 1,2 and all 5 Chest elements	Yes
	HNS NVHead	30cm	Brain	Post Head, Face and 1	Yes
	HNS BrachPlx	40cm	Brachial plexus, C spine	1,2,3 and all 5 Chest elements	Yes
	HNS-CTLTOP	48cm	Largest FoV C and T spine	1,2,3 and 4	No
	HNS-CTLMID	48cm	Largest FoV T and L spine	2,3,4 and 5	No
	HNS CTLBOT	48cm	Largest FoV T and L spine	3,4,5 and 6	No
	HNS-CTL123	40cm	Large FoV C and T spine	1,2 and 3	No
	HNS-CTL234	40cm	Large FoV T spine	2,3 and 4	No
	HNS CTL345	40cm	Large FoV T and L spine	3,4 and 5	No
	HNS CTL456	40cm	Large FoV T and L spine	4,5 and 6	No
	HNS-CTL12	28cm	Smallest FoV C spine	1 and 2	No
	HNS-CTL23	28cm	Smallest FoV Upper T spine	2 and 3	No
	HNS CTL34	28cm	Smallest FoV Mid/Low T spine	3 and 4	No
	HNS CTL45	28cm	Smallest FoV Lower T/L spine	4 and 5	No
	HNS CTL56	28cm	Smallest FoV L spine	5 and 6	No

Table 3-1 Coil Configurations (Continued)

Components Connected	Configurations Available	Max. FoV	Best Use	Element Groups Used	Asset
Posterior Head, Horseshoe, TL Spine and Chest	HNS BrachPlex	40cm	Brachial plexus, C spine	1,2,3 and all 5 Chest elements	Yes
	HNS_CTLTOP	48cm	Largest FoV C and T spine	1,2,3 and 4	No
	HNS_CTLMID	48cm	Largest FoV T and L spine	1,2,3 and 4	No
	HNS CTLBOT	48cm	Largest FoV T and L spine	3,4,5 and 6	No
	HNS_CTL123	40cm	Large FoV C and T spine	1,2 and 3	No
	HNS_CTL234	40cm	Large FoV T spine	2,3 and 4	No
	HNS CTL345	40cm	Large FoV T and L spine	3,4 and 5	No
	HNS CTL456	40cm	Large FoV T and L spine	4,5 and 6	No
	HNS_CTL12	28cm	Smallest FoV C spine	1 and 2	No
	HNS_CTL23	28cm	Smallest FoV Upper T spine	2 and 3	No
	HNS CTL34	28cm	Smallest FoV Mid/Low T spine	3 and 4	No
	HNS CTL45	28cm	Smallest FoV Lower T/L spine	4 and 5	No
	HNS CTL56	28cm	Smallest FoV L spine	5 and 6	No
TL only	HNS CTLBOT	48cm	Largest FoV T and L spine	3,4,5 and 6	No
	HNS CTL345	40cm	Large FoV T and L spine	3,4 and 5	No
	HNS CTL456	40cm	Large FoV T and L spine	4,5 and 6	No
	HNS CTL34	28cm	Smallest FoV Lower T spine	3 and 4	No
	HNS CTL45	28cm	Smallest FoV T/L spine	4 and 5	No
	HNS CTL56	28cm	Smallest FoV L spine	5 and 6	No

Selecting the Correct Configuration

Coil configuration nomenclature is based on the coil components currently connected to the MRI system. For example, with spine configurations, a dash in the configuration name indicates that, along with other components, the Chest component is currently connected to the MR system; an underscore in the name indicates that, along with other components, the Horseshoe **and** Chest components are connected to the system. See Table 3-2 for configuration file naming conventions.

Table 3-2 Configuration File Naming Conventions

Character	Configuration Name	Posterior Head	Face	TL Spine	Chest	Horseshoe
"p"	HNSp CTL12	Yes	Yes	Optional	No	No
"(-)" Dash	HNS-CTL12	Yes	Yes	Optional	Yes	No
"_" Underscore	HNS_CTL12	Yes	No	Optional	Yes	Yes

There are 39 different coil configurations available with the Head, Neck and Spine Coils. The correct selection for your study depends on the following:

- Components connected
- Type of examination (multiple exams on the same patient)
- Desired FoV
- Patient acceptance (when claustrophobia/anxiety prevents using Face component)
- Image quality (if the desired image quality is not realized, an alternate configuration may be chosen)

NOTE: Selecting **[Currently Connected]** from the Coil Selection list narrows the list of usable coil configurations. GE recommends using the Currently Connected feature before selecting configurations.

Some configurations are only compatible when certain coil components are plugged into the system. If the configuration you want is not listed in the Coil Configuration screen, it may be necessary to add or remove coil components before your desired configuration is selectable.

Precise patient positioning and proper FoV will yield optimum results. Improper use of the coil is the major cause of image artifacts.

- Refer to Table 3.1 for configurations optimized for ASSET applications.
- PROPELLER is compatible only with HNS Head and HNS NVHead configurations.
- Gradient Coil Zoom mode is not recommended when using HNS CTL45 and HNS CTL456.

Using Configurations

Please refer to the following section for detailed information concerning the 39 different HNS Coil configurations. The coil components are listed, along with the activated element groups, for each available configuration.

Landmarks are indicated for each configuration. Use these as a guideline when positioning your patients.

The raised laser mark on the Anterior Face component is used for landmarking. The TL Spine component has raised laser marks positioned between the numbered groups identifying the beginning and end of each element group.



CAUTION:

Ensure that no hair or fabric is caught between the components. If the components are not properly connected, oscillation artifacts will appear on the image. If artifacts occur, reseat the components and repeat the scan.

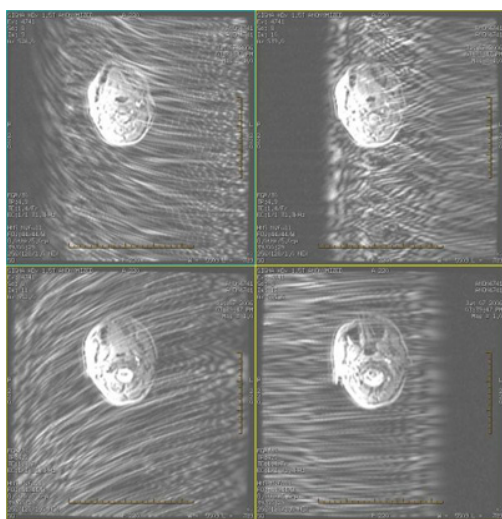
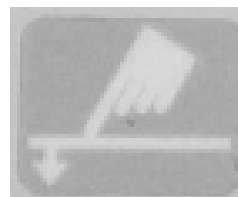


Figure 3-16 Oscillation artifact



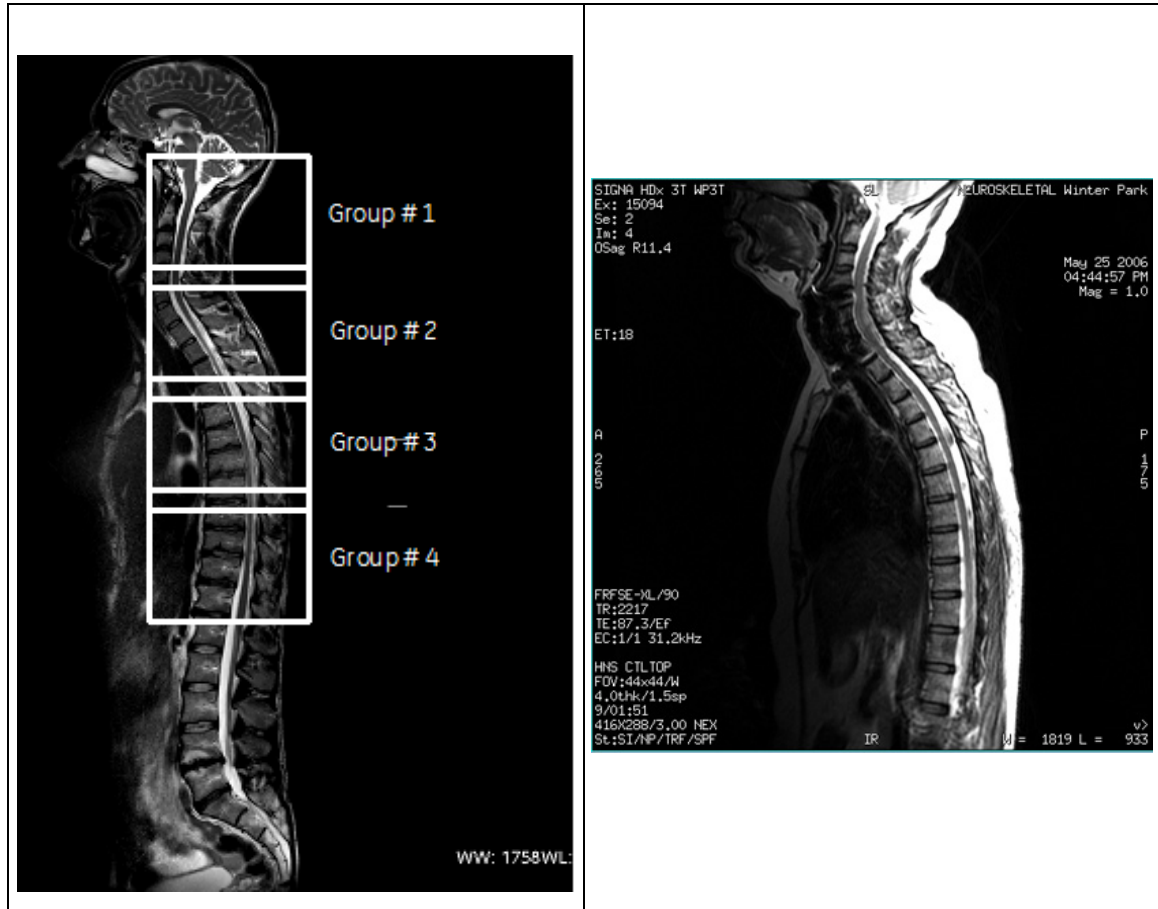
CAUTION:

This label, affixed to the Anterior Face, Horseshoe and Chest connector informs the user to firmly press down on all coil components to ensure that they are fully engaged and connected. Failure to do so may cause artifacts and decrease image quality. Refer to Figure 3-16.

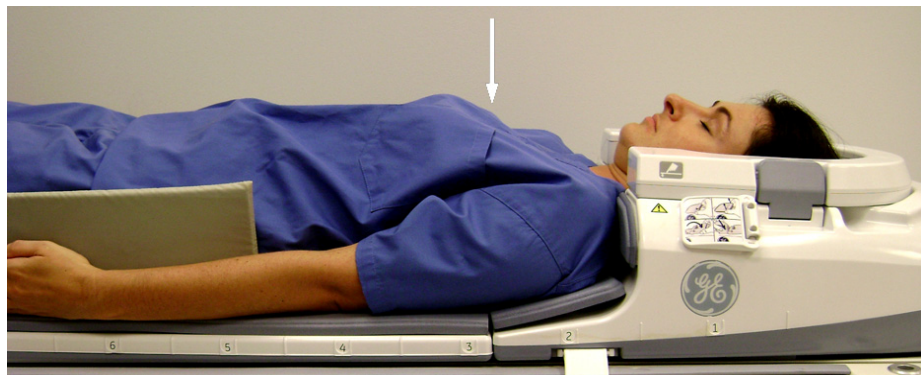


HNS CTLTOP

Components:	Posterior Head, Horseshoe, TL Spine
Element Groups:	1, 2, 3, and 4 (48cm FoV)



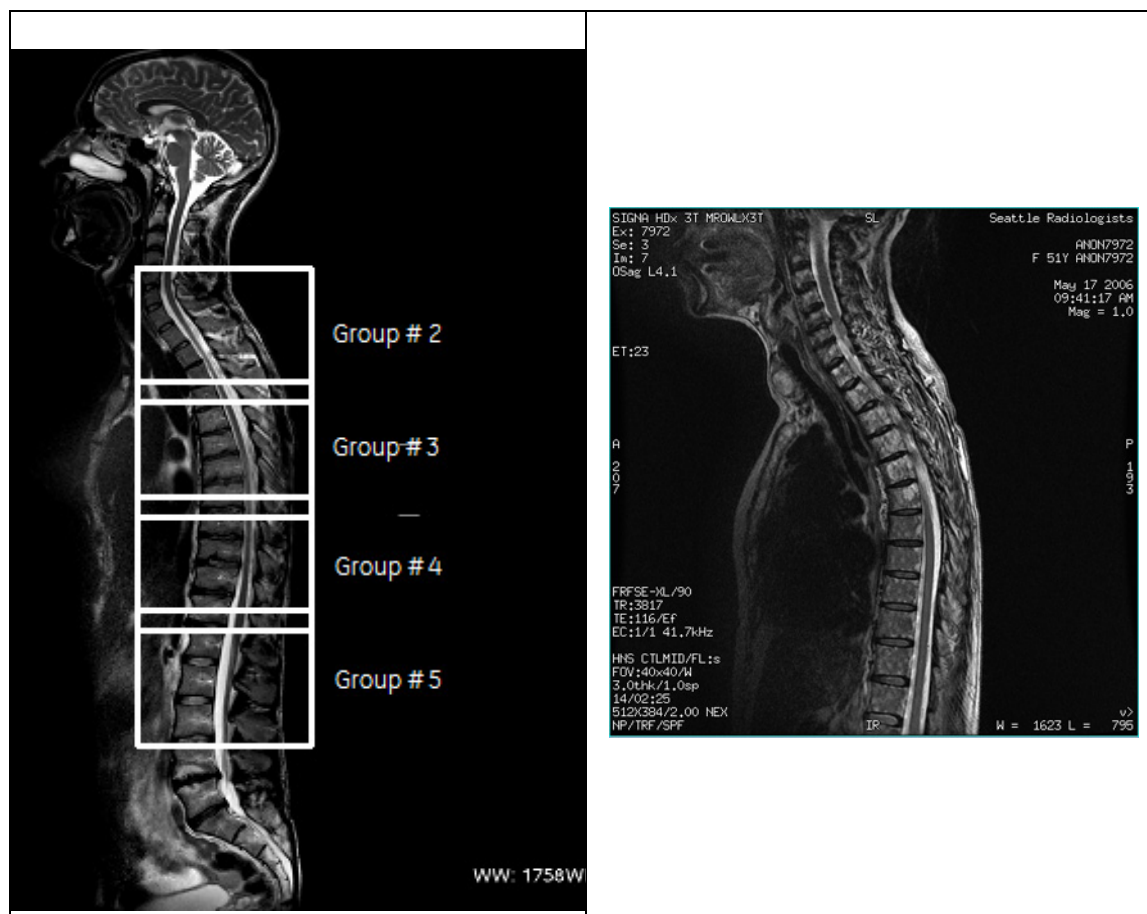
Landmarking:



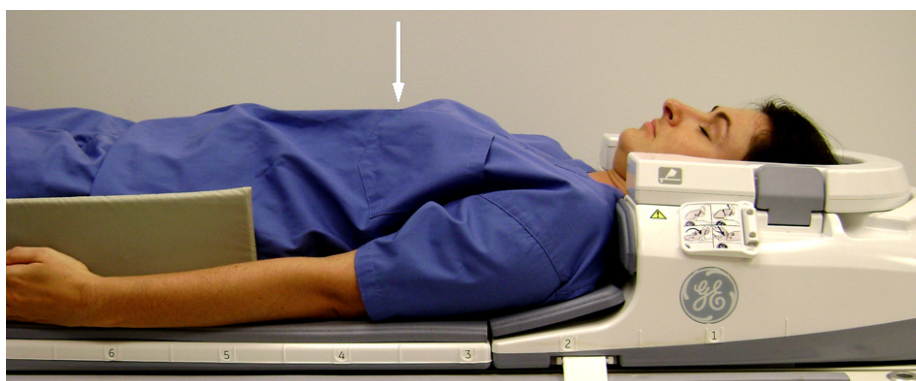
Best Use:	Largest FoV Cervical and Thoracic spine.
------------------	--

HNS CTLMID

Components:	Posterior Head, Horseshoe, TL Spine
Element Groups:	2, 3, 4, and 5 (48cm FoV)



Landmarking:



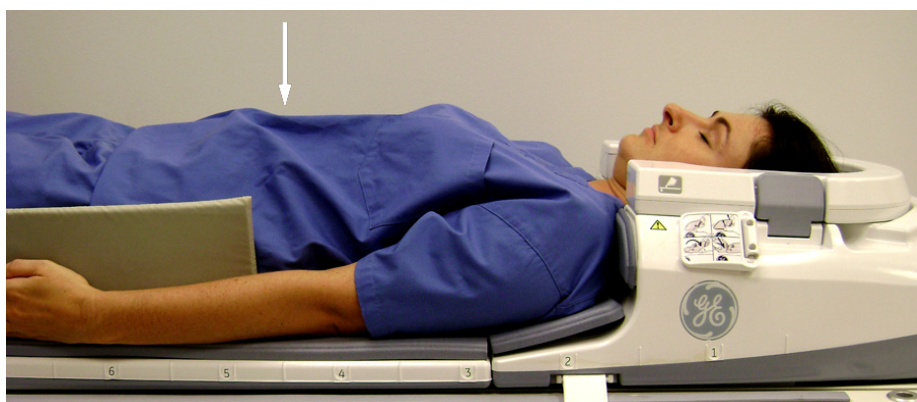
Best Use:	Largest FoV Thoracic and Lumbar spine.
------------------	--

HNS CTLBOT

Components:	TL Spine (with or without Posterior Head and Horseshoe)
Element Groups:	3, 4, 5 and 6 (48cm FoV)



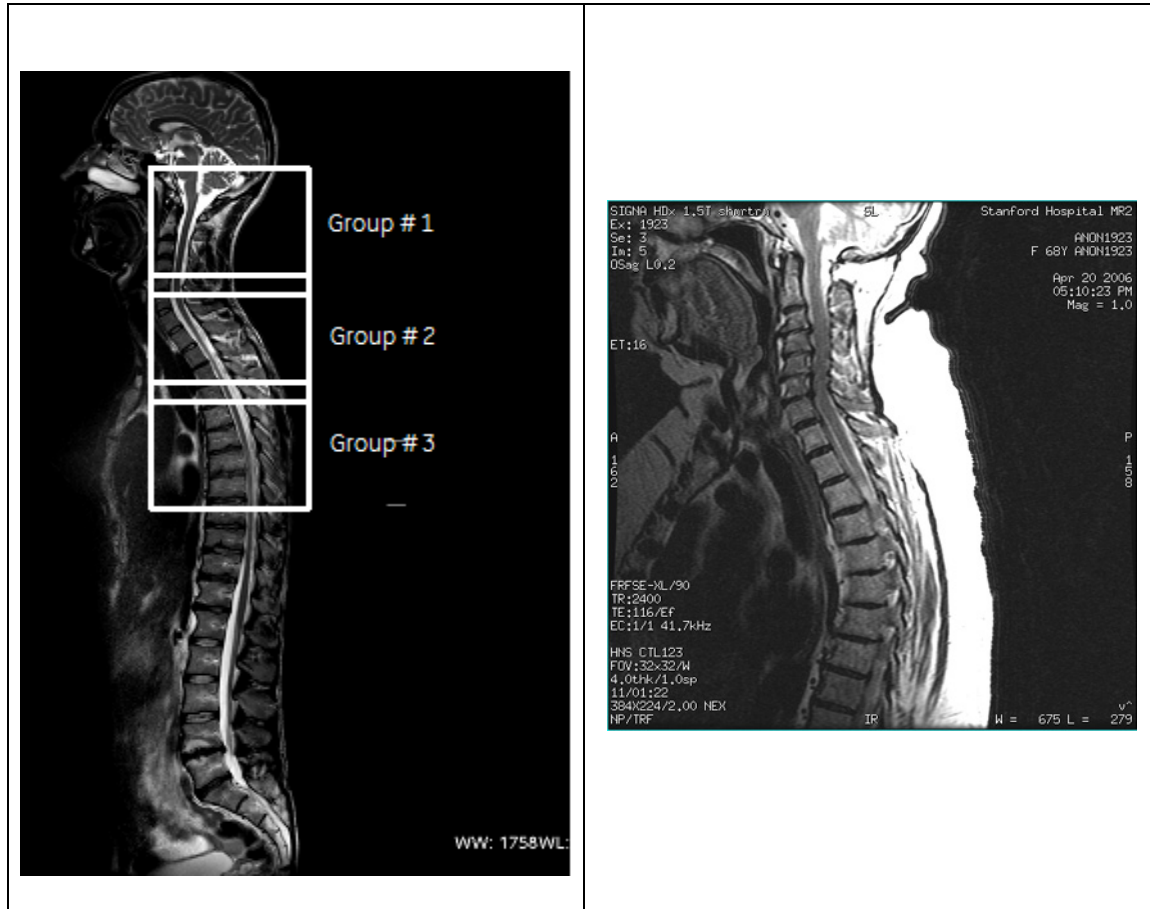
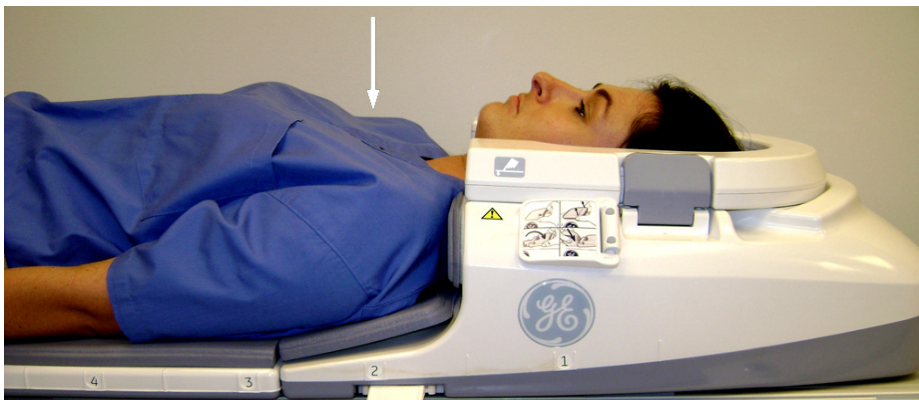
Landmarking:



Best Use:	Largest FoV Thoracic and Lumbar spine.
------------------	--

HNS CTL123

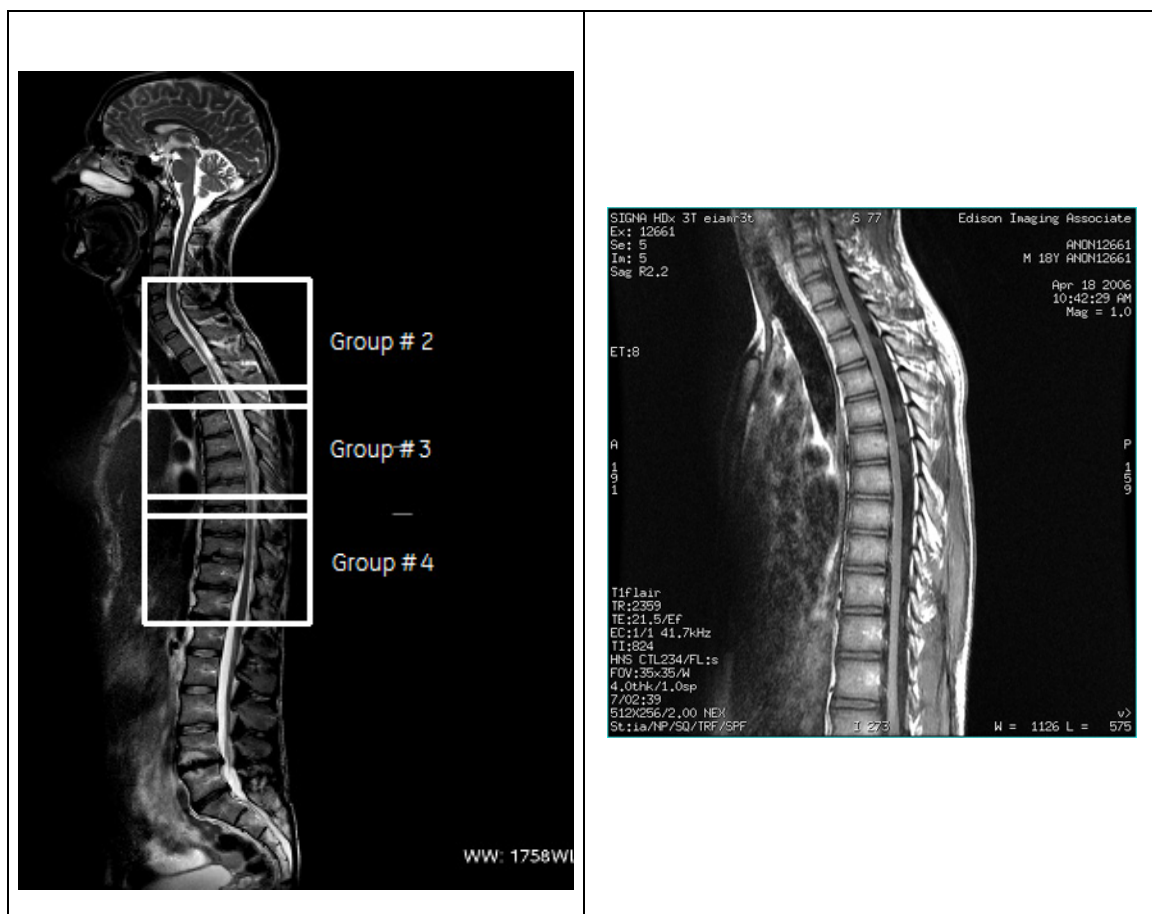
Components:	Posterior Head, Horseshoe, TL Spine
Element Groups:	1, 2 and 3 (40cm FoV)

**Landmarking:**

Best Use:	Large FoV Cervical and Thoracic spine
------------------	---------------------------------------

HNS CTL234

Components:	Posterior Head, Horseshoe and TL Spine
Element Groups:	2, 3 and 4 (40cm FoV)



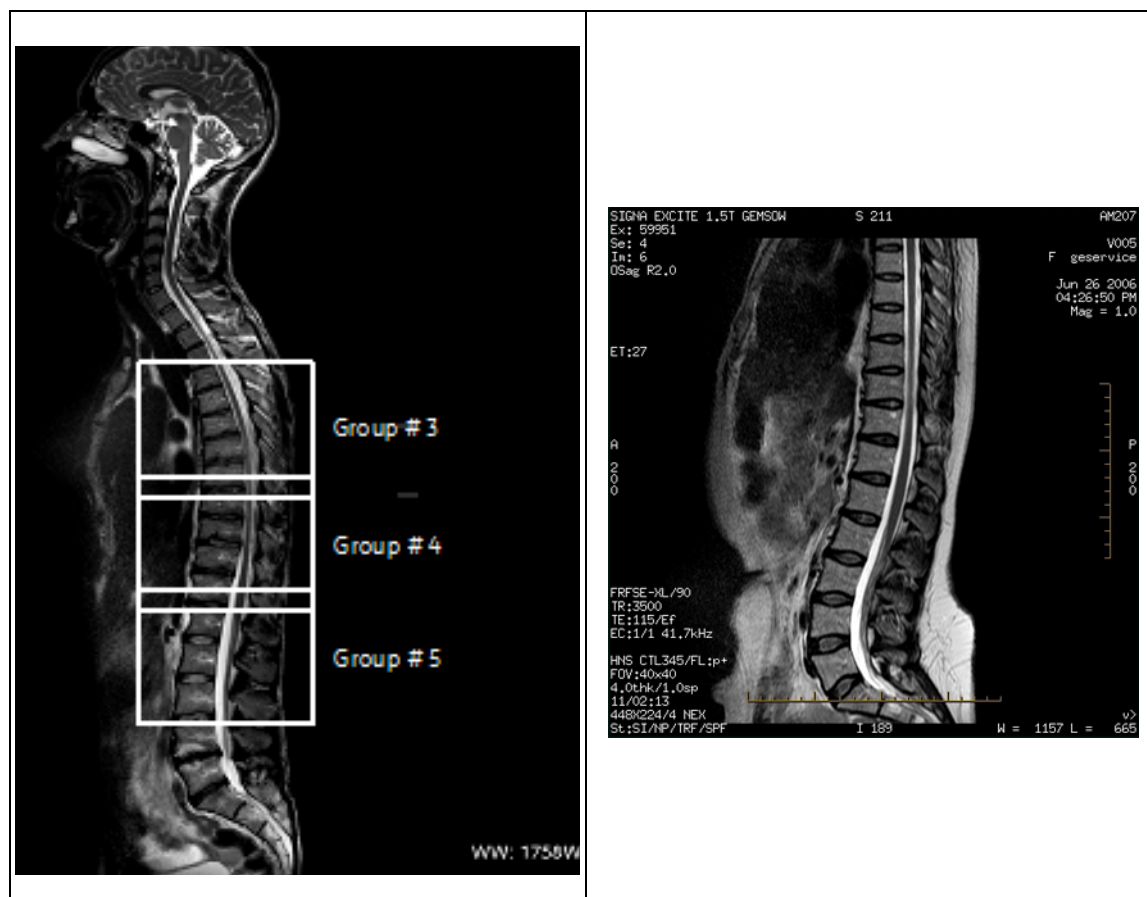
Landmarking:



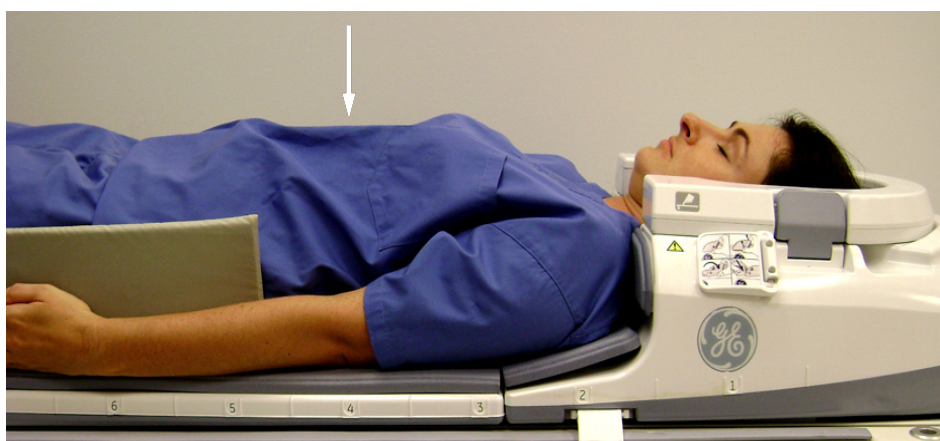
Best Use:	Large FoV Thoracic spine.
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HNS CTL345

Components:	TL Spine (with or without Posterior Head and Horseshoe)
Element Groups:	3, 4 and 5 (40cm FoV)



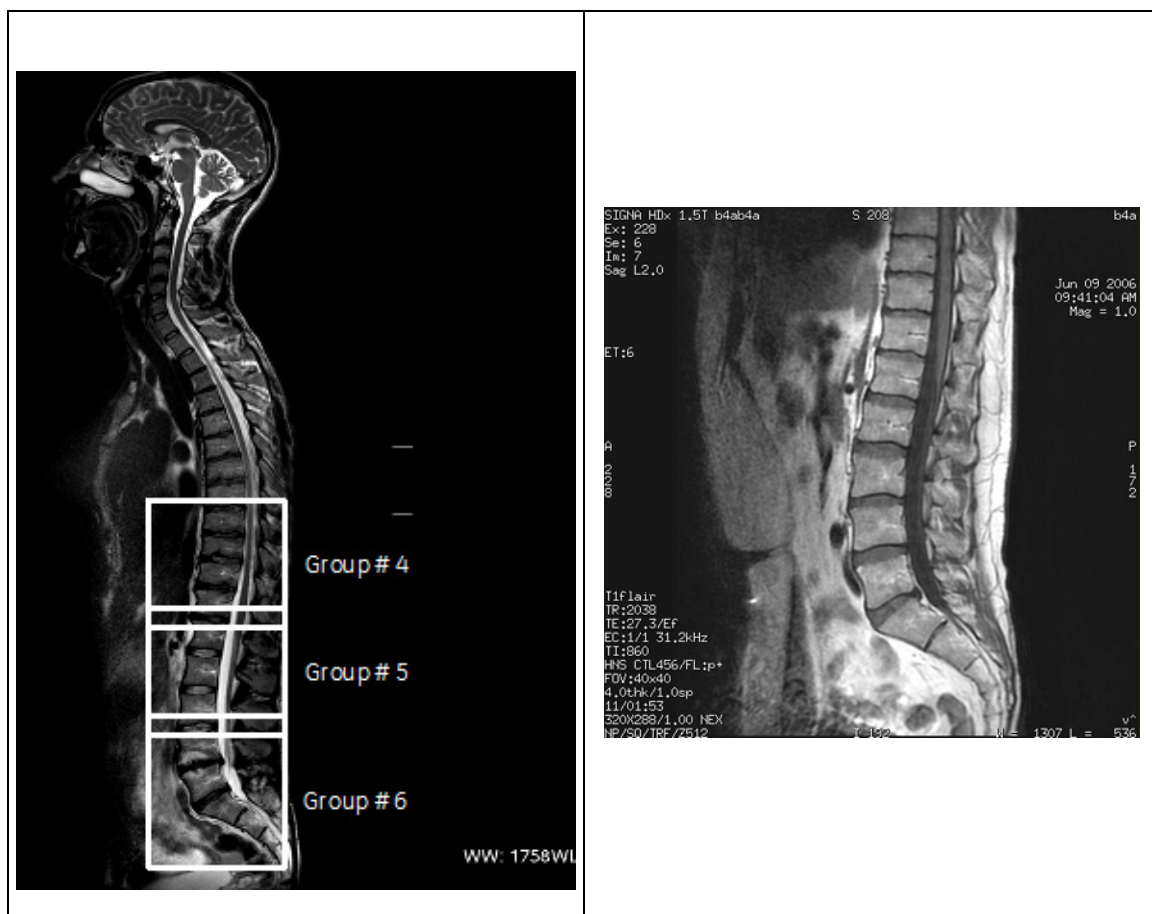
Landmarking:



Best Use:	Large FoV Thoracic/Lumbar spine.
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HNS CTL456

Components:	TL Spine (with or without Posterior Head and Horseshoe)
Element Groups:	4, 5 and 6 (40cm FoV)



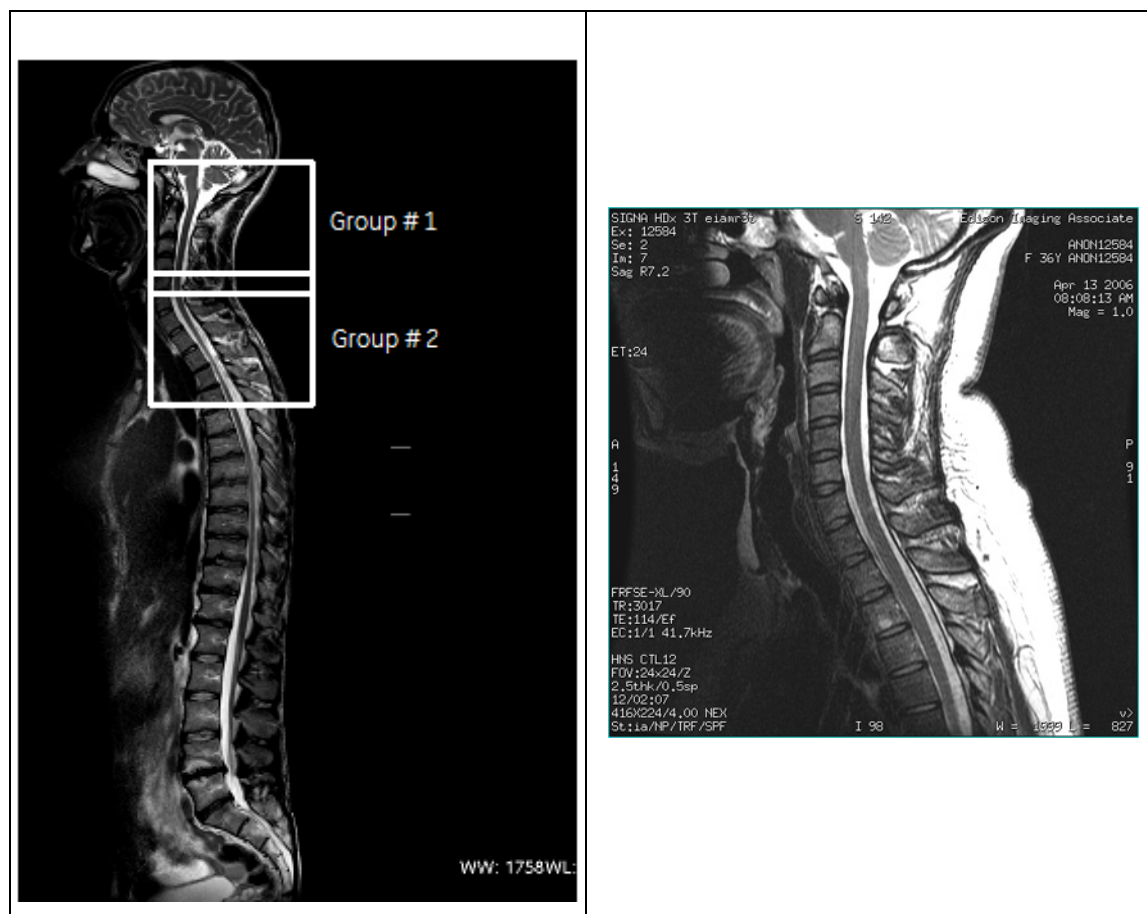
Landmarking:



Best Use:	Largest FoV Thoracic/Lumbar spine.
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HNS CTL12

Components:	Posterior Head and Horseshoe (with or without TL Spine)
Element Groups:	1 and 2 (28cm FoV)



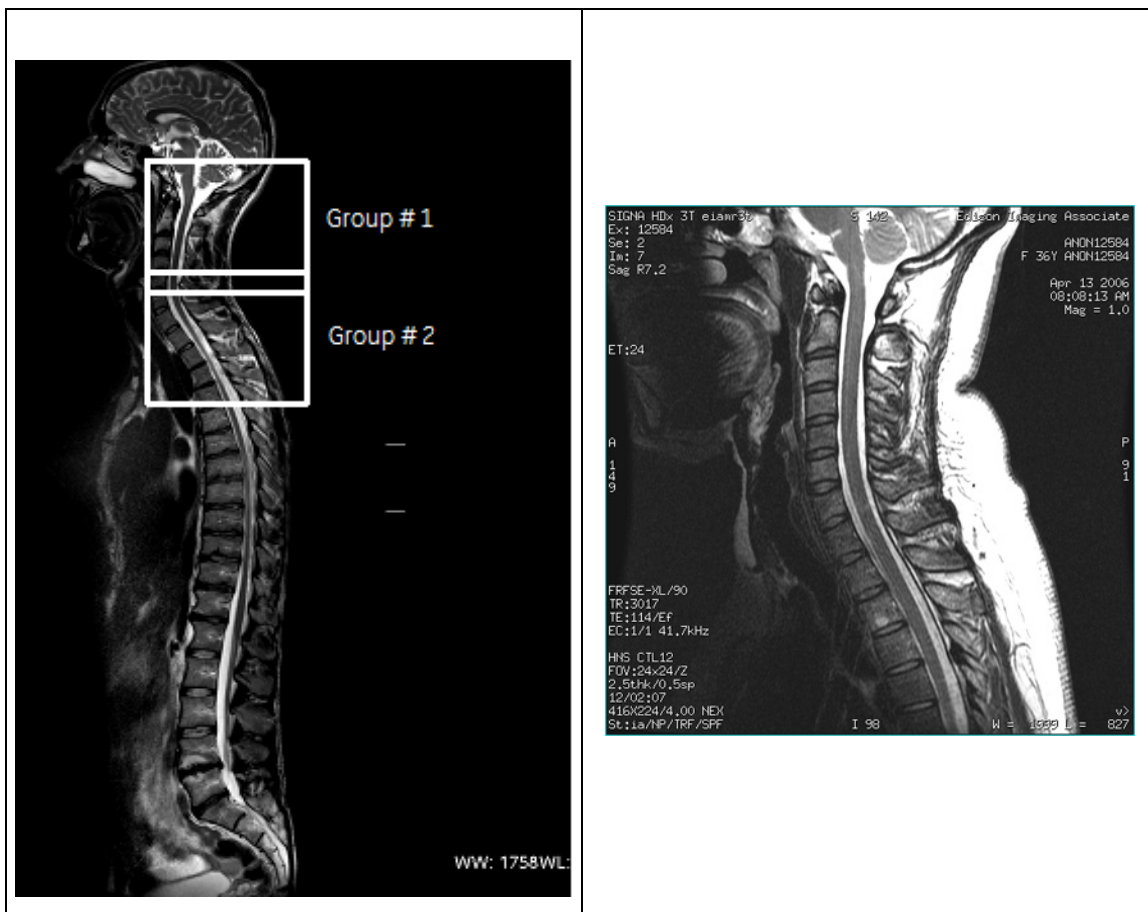
Landmarking:



Best Use:	Smallest FoV Cervical Spine
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HNSp CTL12

Components:	Posterior Head and Face (with or without TL Spine)
Element Groups:	1 and 2 (28cm FoV)



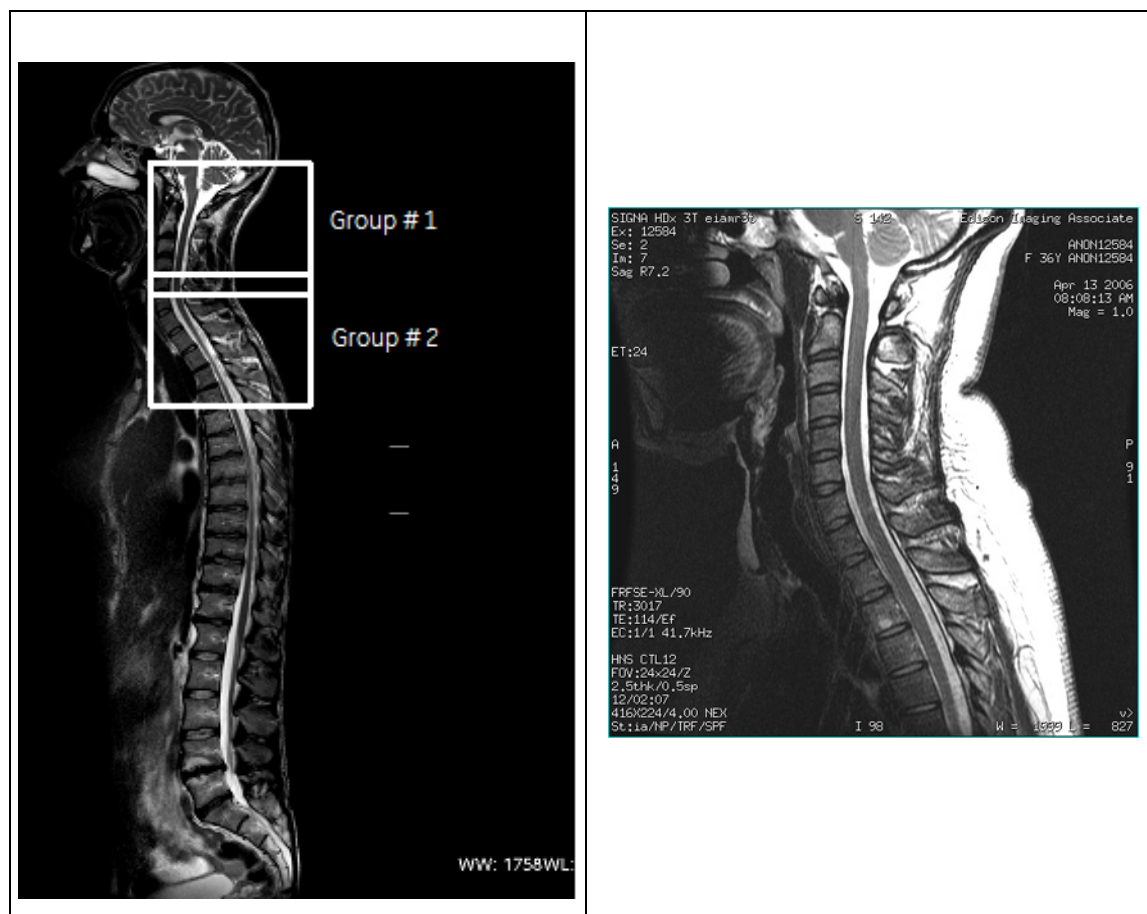
Landmarking:



Best Use:	Smallest FoV Cervical spine
------------------	-----------------------------

HNS-CTL12

Components:	Posterior Head, Face and Chest (with or without TL Spine)
Element Groups:	1 and 2 (28cm FoV)



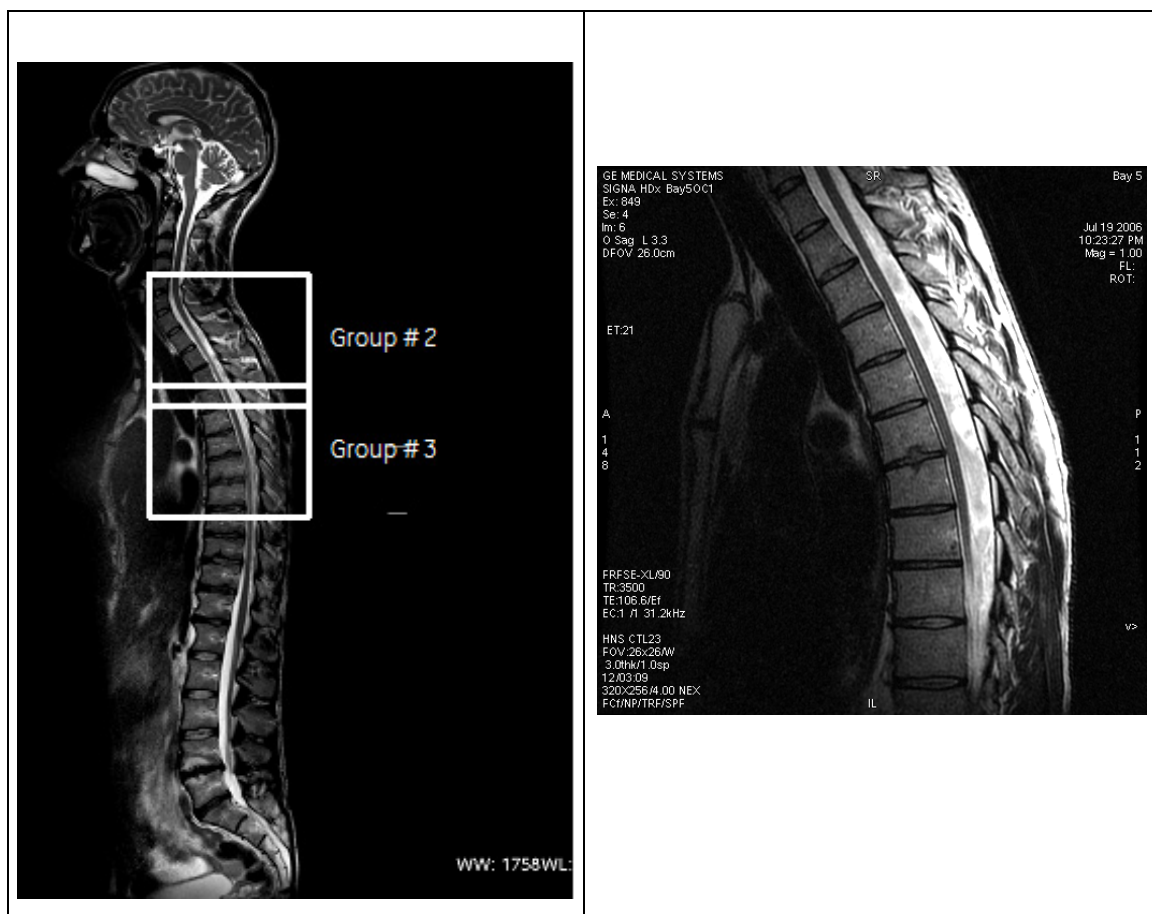
Landmarking:



Best Use:	Smallest FoV Cervical spine
------------------	-----------------------------

HNS CTL23

Components:	Posterior Head, Horseshoe and TL Spine
Element Groups:	2 and 3 (28cm FoV)



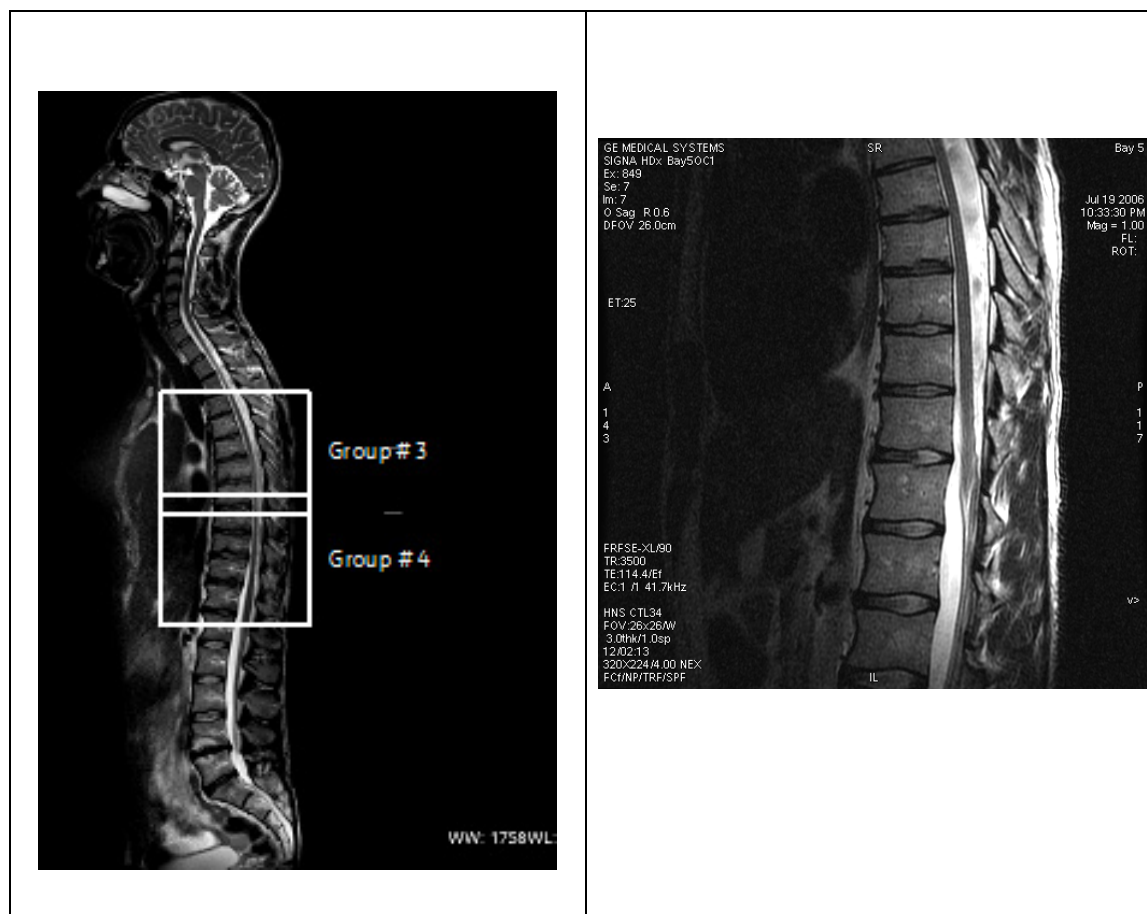
Landmarking:



Best Use:	Smallest FoV Upper Thoracic spine
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HNS CTL34

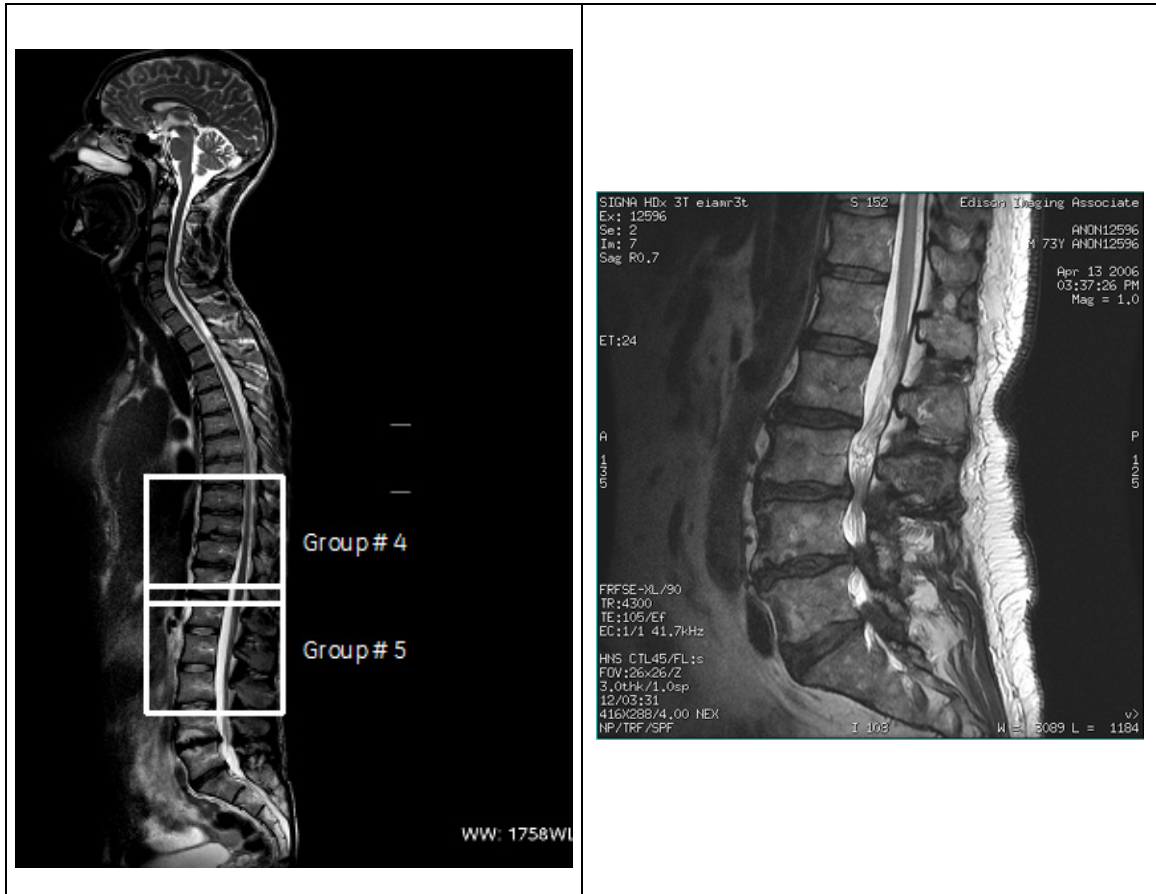
Components:	TL Spine (with or without Posterior Head and Horseshoe)
Element Groups:	3 and 4 (28cm FoV)

**Landmarking:**

Best Use:	Smallest FoV Lower Thoracic spine
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HNS CTL45

Components:	TL Spine (with or without Posterior Head and Horseshoe)
Element Groups:	4 and 5 (28cm FoV)



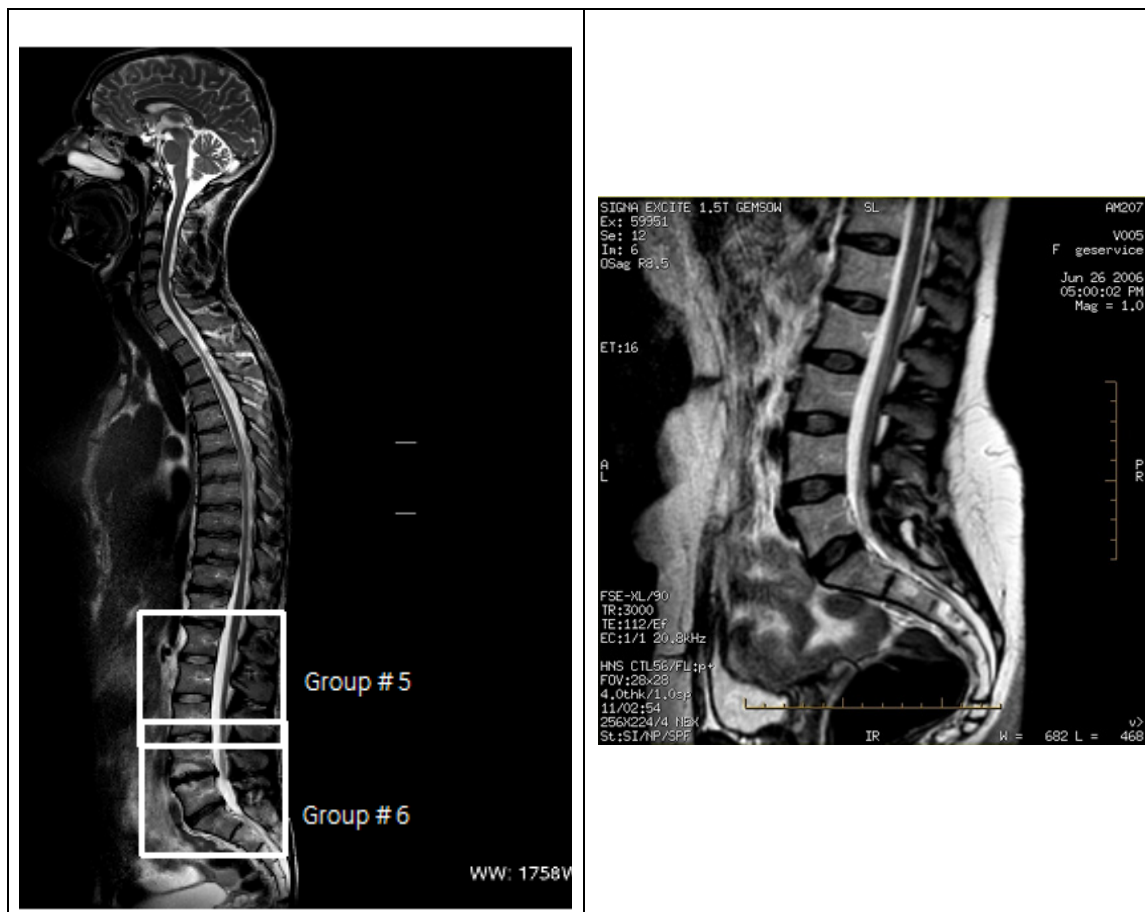
Landmarking:



Best Use:	Smallest FoV Thoracic/Lumbar spine
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HNS CTL56

Components:	TL Spine (with or without Posterior Head and Horseshoe)
Element Groups:	5 and 6 (28cm FoV)



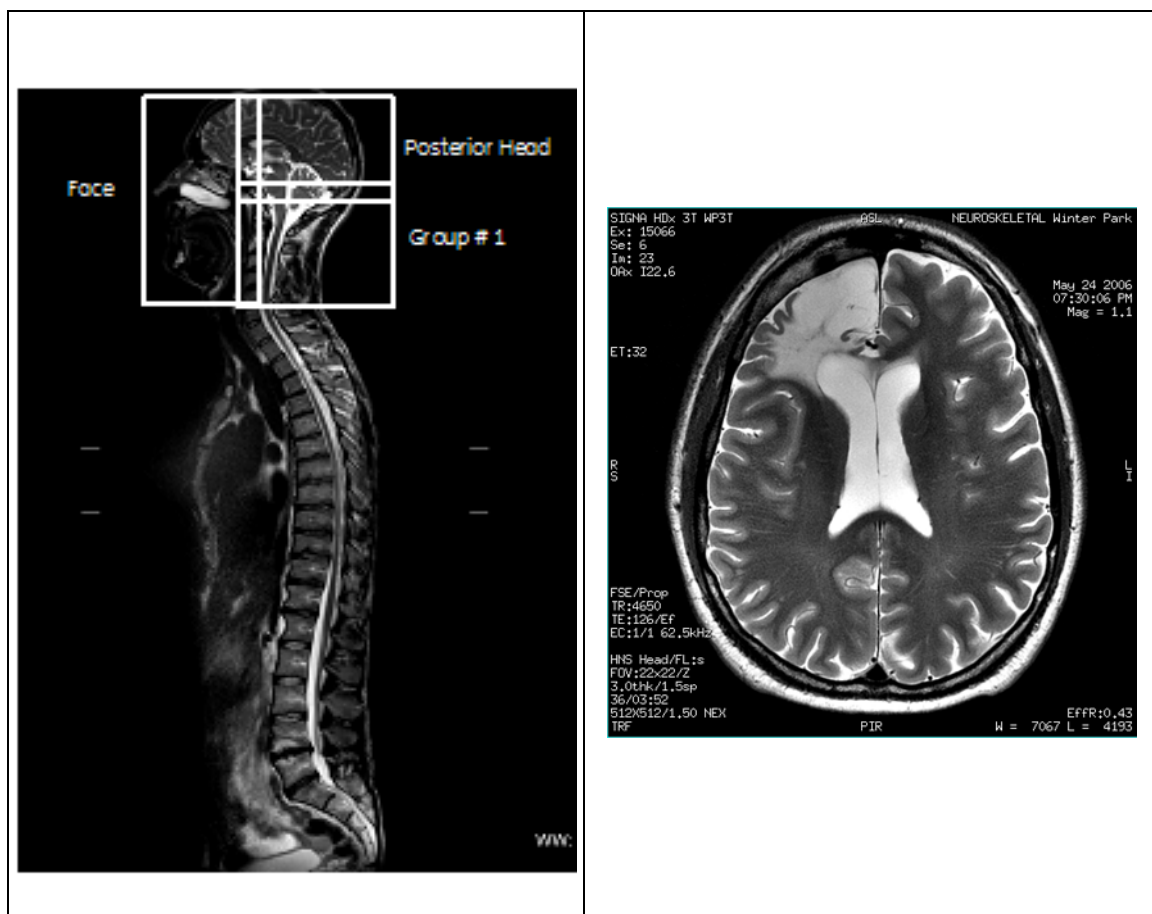
Landmarking:



Best Use:	Smallest FoV Lumbar spine.
------------------	----------------------------

HNS Head

Components:	Posterior Head and Face (with or without TL)
Element Groups:	Post Head, Face, and 1 (30cm FoV)



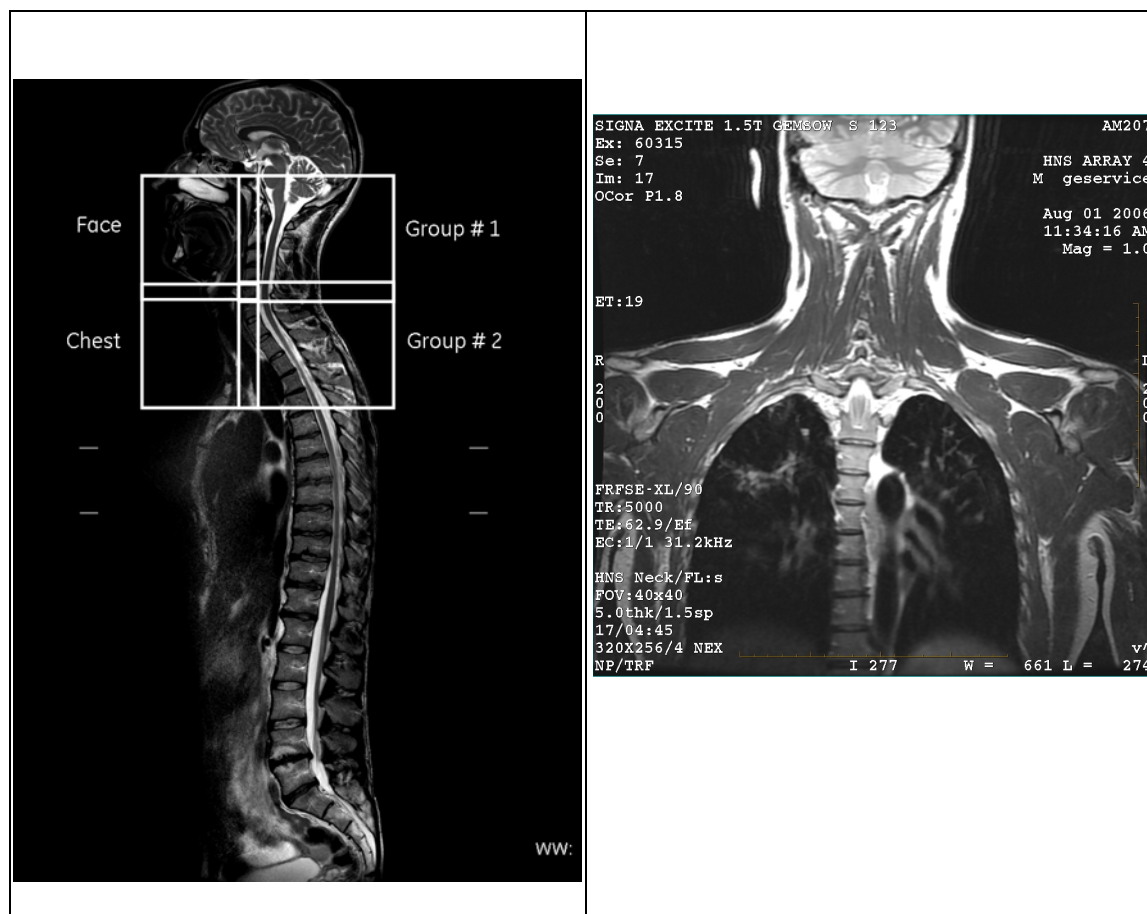
Landmarking:



Best Use:	Brain
------------------	-------

HNS Neck

Components:	Posterior Head, Face and Chest (with or without TL)
Element Groups:	Face, 1, 2, and 2 Chest elements (30cm FoV)



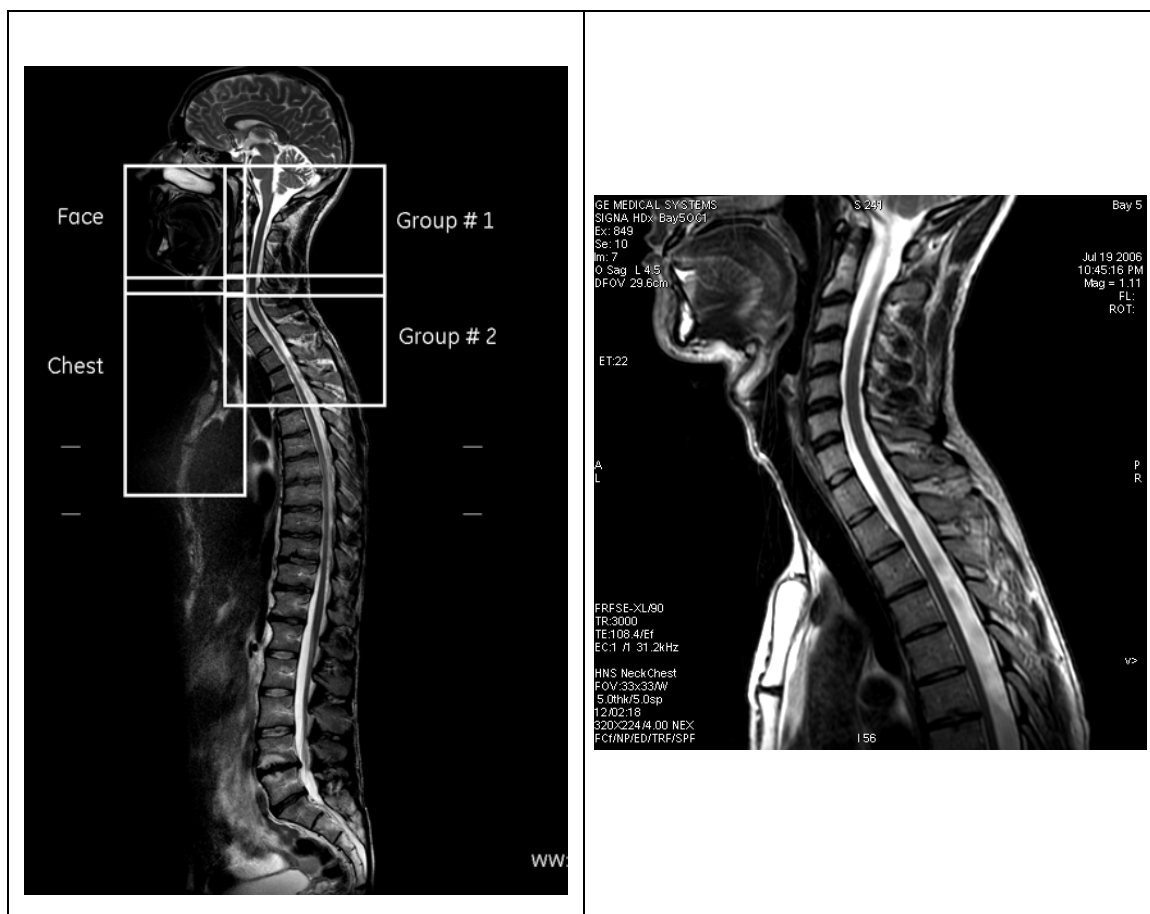
Landmarking:



Best Use:	Smallest FoV Cervical spine, Carotids, Soft Tissue neck.
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HNS NeckChest

Components:	Posterior Head, Face and Chest (with or without TL)
Element Groups:	Face, 1, 2, and 5 Chest elements (35cm FoV)



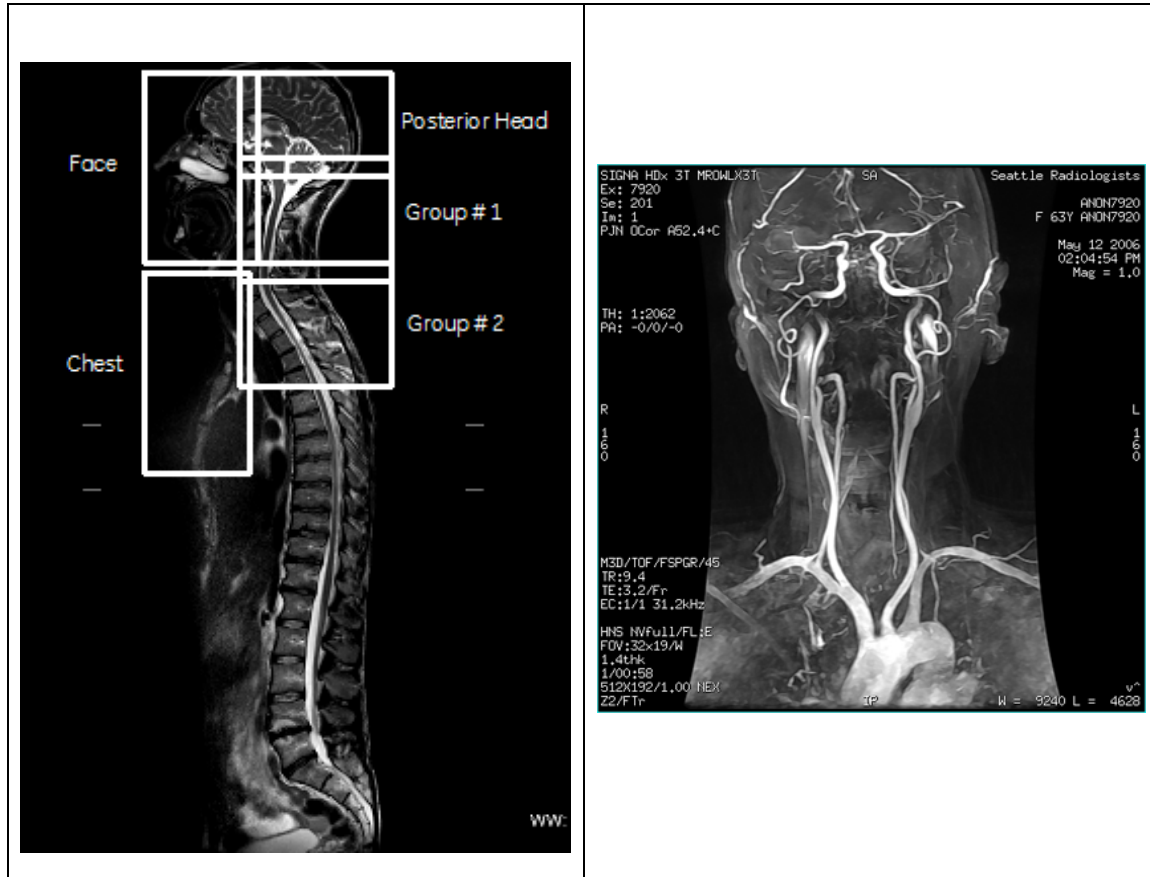
Landmarking:



Best Use:	Large FoV Cervical spine, Carotids, Soft Tissue neck.
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HNS NVFull

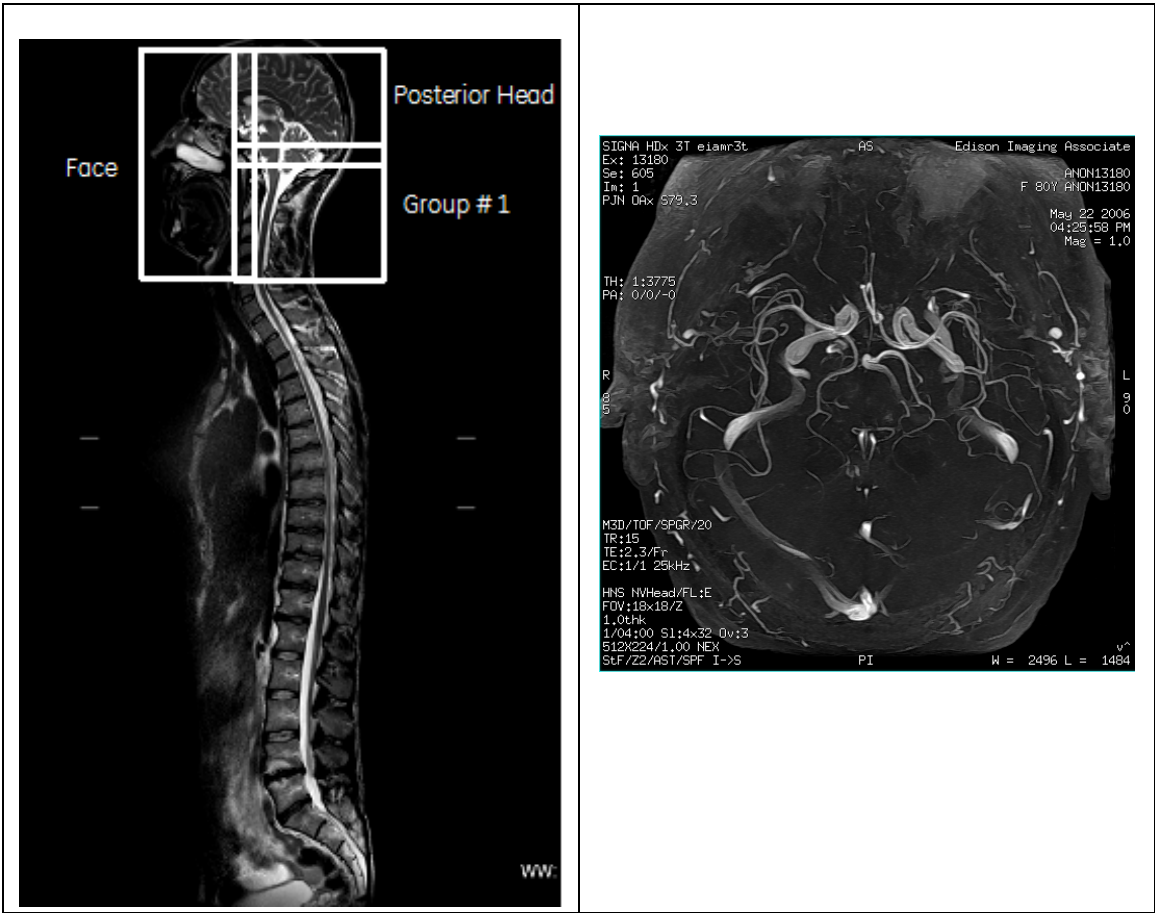
Components:	Posterior Head, Face and Chest (with or without TL)
Element Groups:	Post Head, Face, 1, 2, and all 5 Chest elements (48cm FoV)

**Landmarking:**

Best Use:	Largest FoV Carotids, Cervical spine and Soft Tissue neck.
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HNS NVHead

Components:	Posterior Head, Face and Chest (with or without TL)
Element Groups:	Post Head, Face and 1 (30cm FoV)



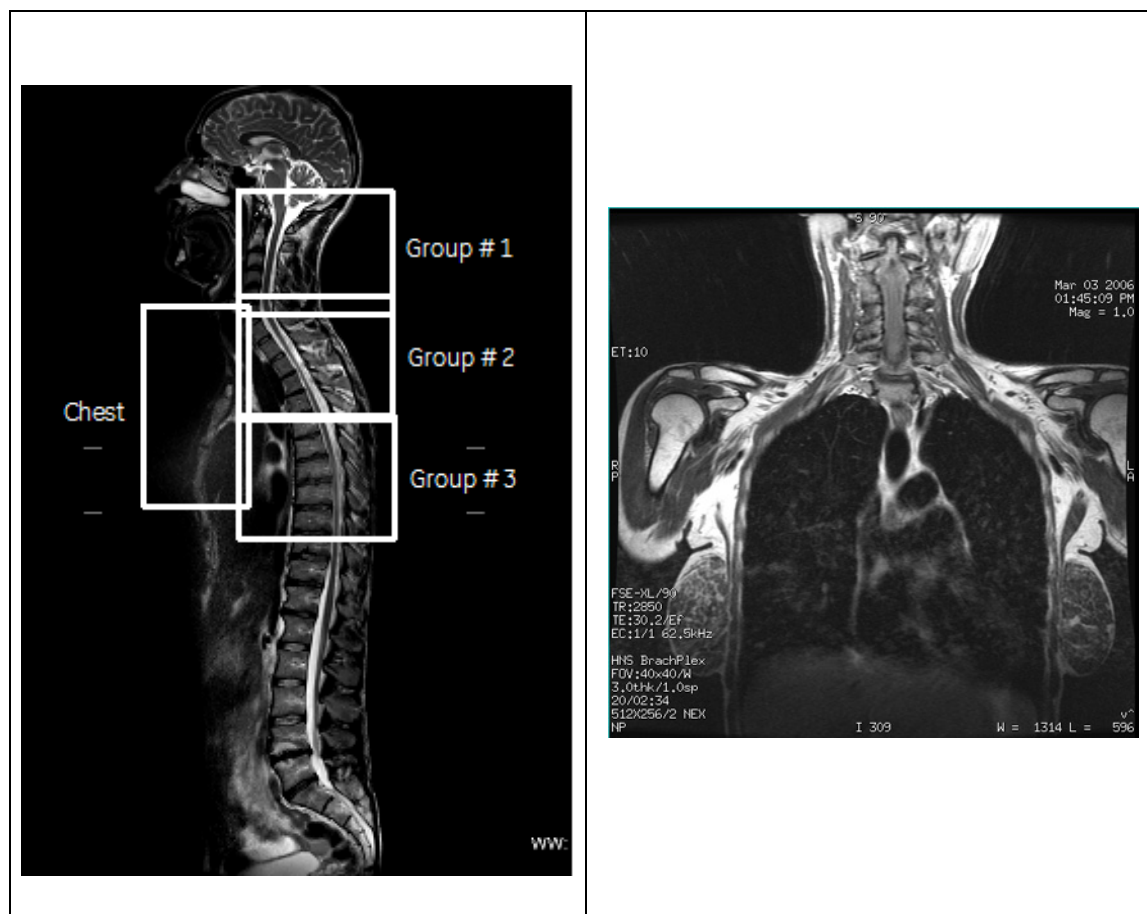
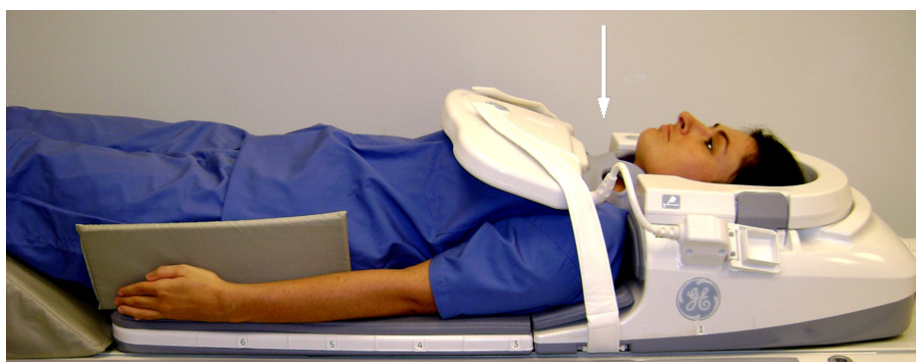
Landmarking:



Best Use:	Brain
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HNS BrachPlex

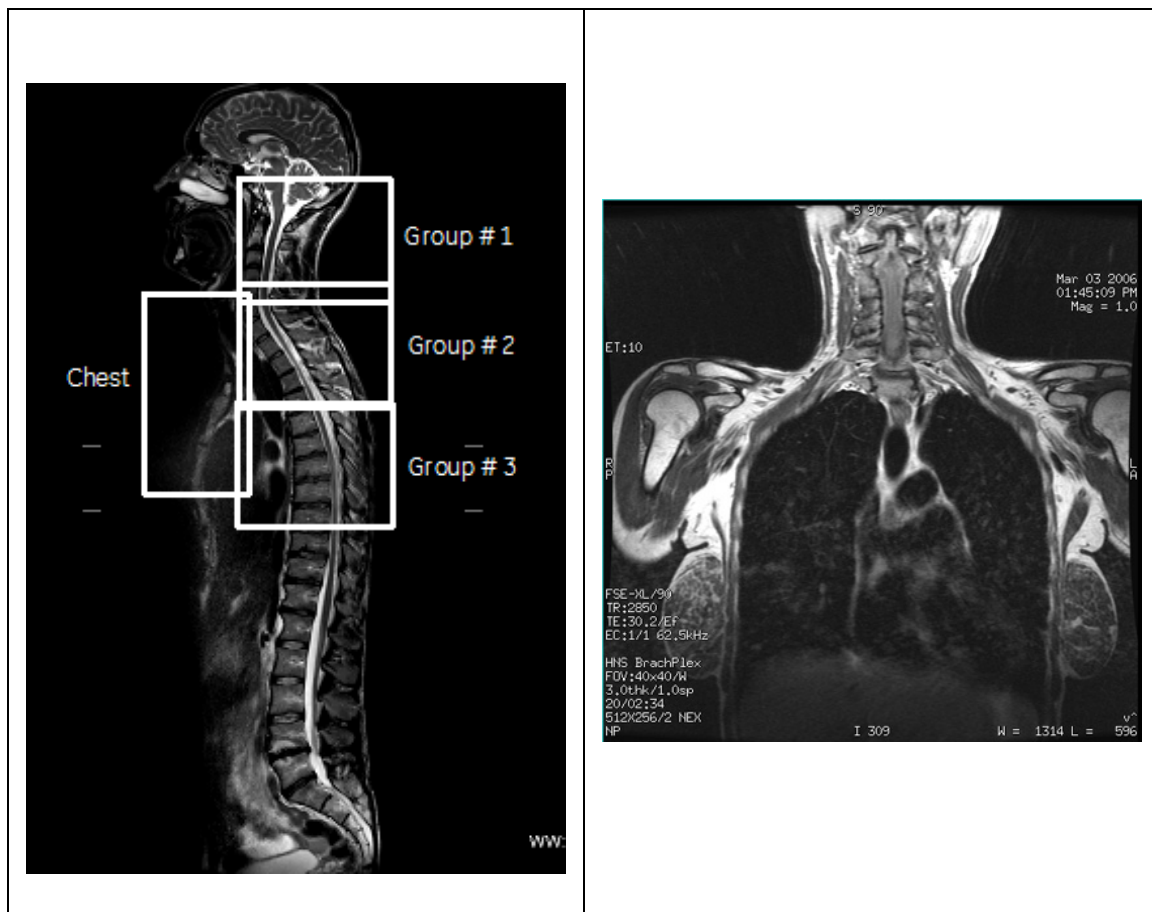
Components:	Posterior head, Horseshoe, Chest and TL
Element Groups:	1, 2, 3 and all 5 Chest elements (40cm FoV)

**Landmarking:**

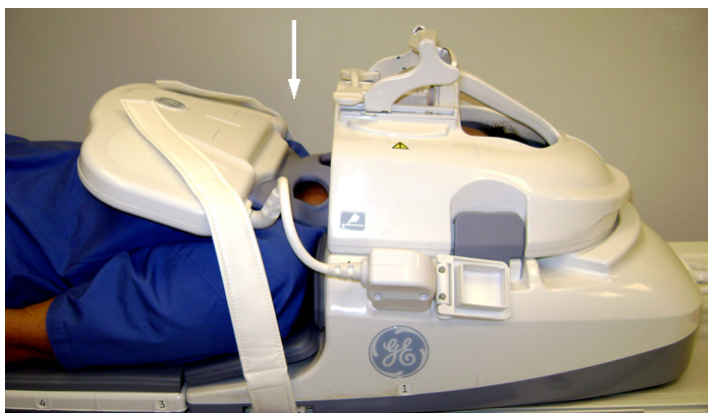
Best Use:	Brachial Plexus, Cervical spine.
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HNS BrachPlx

Components:	Posterior Head, Face, Chest, and TL
Element Groups:	1, 2, 3 and all 5 Chest elements (40cm FoV)



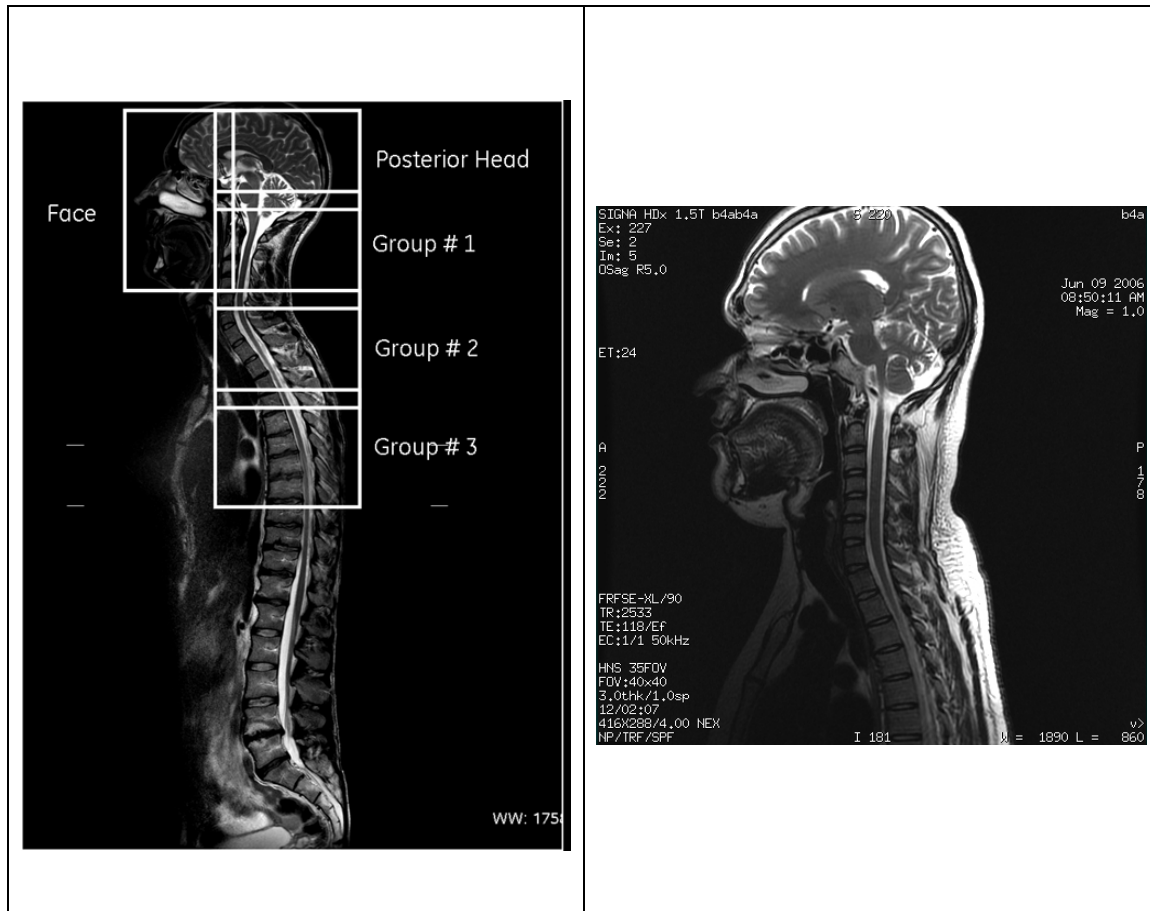
Landmarking:



Best Use:	Brachial Plexus, Cervical Spine.
------------------	----------------------------------

HNS 35FOV

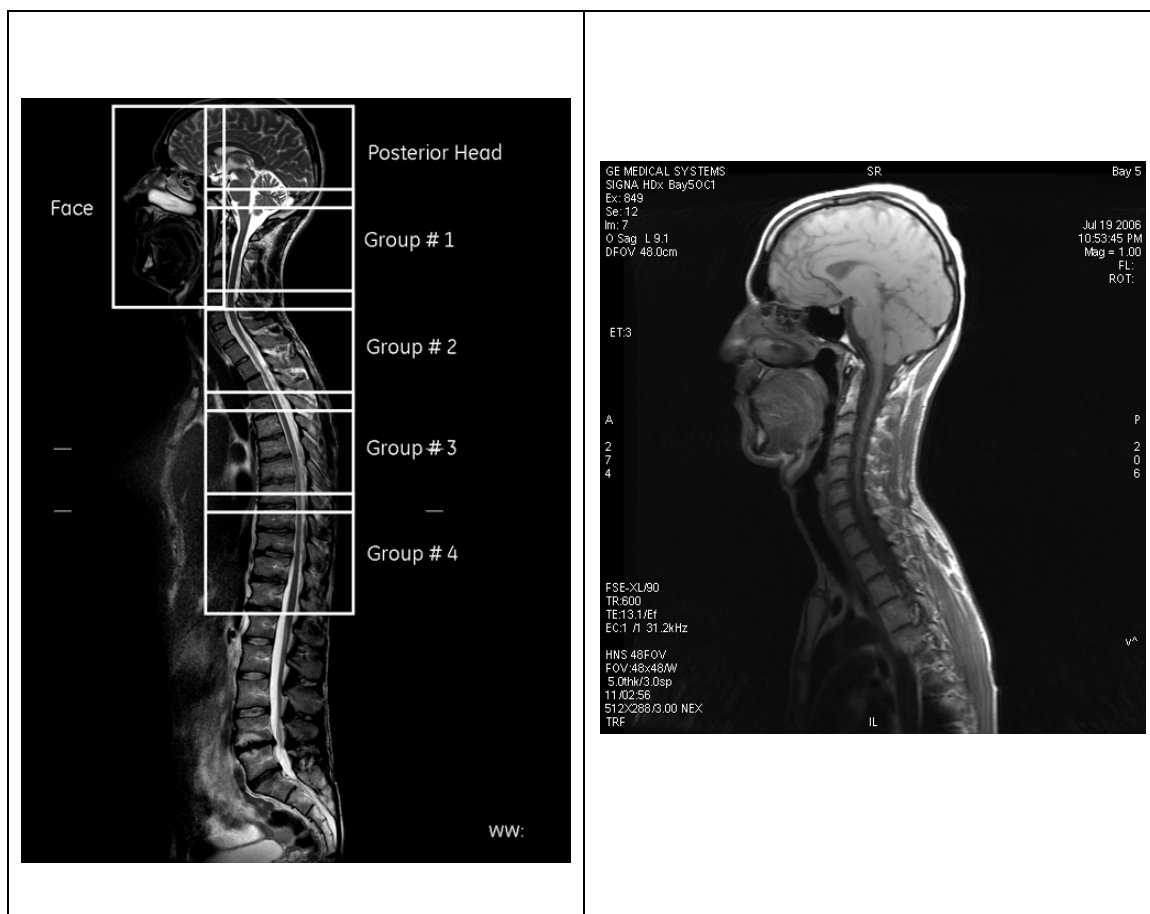
Components:	Posterior Head, Face and TL
Element Groups:	Post Head, Face, 1, 2, and 3 (35-40cm FoV)

**Landmarking:**

Best Use:	Large FoV Brain, Cervical Spine.
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HNS 48FOV

Components:	Posterior Head, Face and TL
Element Groups:	Post Head, Face, 1, 2, 3, and 4 (40-48cm FoV)



Landmarking:



Best Use:	Largest FoV Brain, Cervical spine, Thoracic spine.
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Using the Equivalent Spine Configurations

The previous Section, “Using Configurations,” only shows the scan image, selected element groups and landmark position for the 12 spine configurations that are available when using Posterior Head, Horseshoe and TL spine. For convenience, each spine configuration has 3 equivalent configurations with other coil components connected. The following table shows the equivalent spine configurations. These equivalent configurations use the same element groups and have the same coverage. Refer to the previous section for the landmark information for these equivalent spine configurations.

Table 3-3 Equivalent Spine Configurations

Spine Configurations With Posterior Head and Horseshoe	Equivalent Spine Configurations		
	Posterior Head and Face	Posterior Head, Face and Chest	Posterior Head, Horseshoe and Chest
HNS CTL12	HNSp CTL12	HNS-CTL12	HNS_CTL12
Spine Configurations With Posterior Head, Horseshoe and TL Spine	Equivalent Spine Configurations		
	Posterior Head, Face and TL Spine	Posterior Head, Face, Chest and TL Spine	Posterior Head, Horseshoe, Chest and TL Spine
HNS CTL23	HNSp CTL23	HNS-CTL23	HNS_CTL23
HNS CTL123	HNSp CTL123	HNS-CTL123	HNS_CTL123
HNS CTL234	HNSp CTL234	HNS-CTL234	HNS_CTL234
HNS CTLTOP	HNSp CTLTOP	HNS-CTLTOP	HNS_CTLTOP
HNS CTLMID	HNSp CTLMID	HNS-CTLMID	HNS_CTLMID

Coil and Patient Positioning



WARNING:

Remove any other coil or unused accessory device from the magnet before using the coil. Patient burns may result from RF coupling of this device with extraneous devices in the magnet.

NOTE: For ease of use, position the patient restraint straps in the desired location on the table before placing the TL Component on the table.

Table 3-4 Coil and Patient Positioning

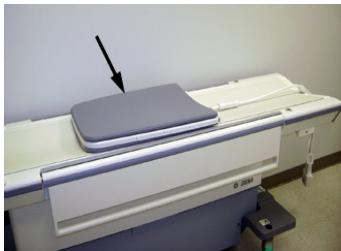
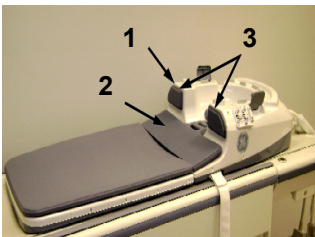
Visual	Description
	1. Place the TL Spine component (arrow) on the table, then place the pad on it. Patient will enter the scanner head first, so be sure cables extend toward the docking end of the table.
	2. Place the Posterior Head component (arrow 1) on the table as shown, making sure it clears the cradle docking mechanism. Interlock the pegs on the bottom with the holes in the TL Spine Component. Place pad on the Posterior Head component as shown (arrow 2). Place the patient pads on the posterior Head component as shown (arrow 3). NOTE: Ensure that the TL cable lies within the cable channel beneath the Posterior Head component. When placed properly, the Posterior Head component will sit evenly on top of the TL cable.
	3. Position the patient in the supine position on the table and lower the head into the Posterior Head component. Make sure that shoulders touch the shoulder rest area (arrow).

Table 3-4 Coil and Patient Positioning (Continued)

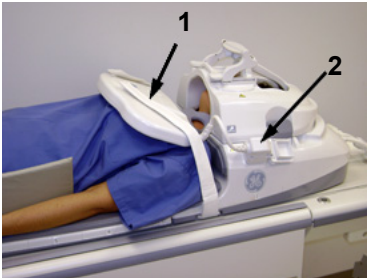
Visual	Description
	4. Place the Face component (arrow 1) over the Posterior Head component. Secure Face component with the blue latches on both sides (arrow 2).  CAUTION: Ensure that no hair or fabric is caught between the components. Failure to comply may cause artifacts and decreased image quality.
	5. Position the Chest component (arrow 1) so that the neck portion fits snugly against the patient's chin. Plug the chest piece connector into the Posterior Head component (arrow 2).  CAUTION: Ensure that the Chest component is plugged securely into the connector on the Posterior Head component. Failure to comply may cause artifacts and decreased image quality.  CAUTION: The Chest component may not fit in the bore on patients with large chests. To avoid injuring the patient, watch carefully as the table moves into the bore, and stop advancing the table if the Chest component comes into contact with the top of the bore.
	6. When scanning only C, T or L spine, it is not necessary to attach the Face component. Instead, the Horseshoe (arrow) may be used. It is positioned much like the Face component and is secured with the blue latches on either side of the Posterior Head component. NOTE: The Face component or the Horseshoe MUST be in place for the coil to function properly when the Posterior Head component is used.  CAUTION: Ensure that the Horseshoe component is completely seated on the Posterior Head component. Failure to comply may cause artifacts and decreased image quality.

Table 3-4 Coil and Patient Positioning (Continued)

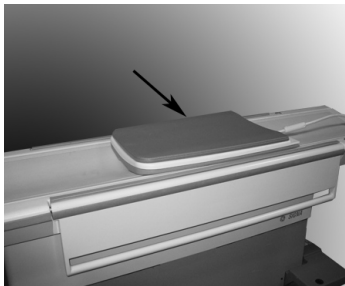
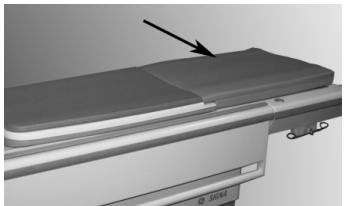
Visual	Description
	7. If using only the TL Spine component of the array, place TL component on the table, making sure cable extends to docking end of table as shown.
	8. Place patient comfort pad over cable as shown. Make sure that cable follows the groove in the underside of the pad. 9. Position patient and then plug the TL connector into port A. If you are using only the TL Spine component, you may plug the TL connector into port B. In general, always use port A. NOTE: The patient may be positioned Head first or Feet first when using the TL Spine component independently.

Table 3-4 Coil and Patient Positioning (Continued)

Visual	Description
 HD Connector  P connector  Legacy Bendix connector  3.0T Legacy Connector	10. There are multiple system coil plug-in configurations. - For a three connector HD coil - Plug the A connector into the A port, the B connector into the B port and the Legacy Bendix into the Bendix port for a 1.5T System or the 3.0T Legacy Connector into the Legacy port for a 3.0T System. - For a two connector HD Coil - Plug the A connector into the A port and the B connector into the B port. - For a P connector coil - Plug the A connector into the A port and the P connector into either Port P1 or P2. Lock the P connector plug by rotating the spindle handle until the Lock symbol is visible. NOTE: Make sure all cables are plugged in and that they do not cross or loop. When cables labeled A or B are plugged in, the green light illuminates.  CAUTION: All coil components must be plugged into the system when they are in the scanner. Leaving components unplugged can damage the coil, or cause harm to the patient. 11. Disconnect coil cables before lowering the patient table.  CAUTION: Unplug the coil before lowering the table. Failure to do so may cause damage to the coil and connector and may render the coil inoperable.

**WARNING:**

Hearing protection is required for all people in the scan room during a scan to prevent hearing impairment. Acoustic levels may exceed 99dB(A). Provide the patient with hearing protection that has a Noise Reduction Rating (NRR) of 29dB or better.

Localizing anatomy

Position the area of interest to the center of the coil elements (based on selected configuration). Please refer to Table 3-1 for detailed coil configurations. Landmark to the center of the coil elements at the region of interest.

Recommended scan protocols

Use the system recommended protocols for scanning with this coil for optimum results and best performance.

Specific choice of protocols is entirely the responsibility of the radiologist and/or technologist.

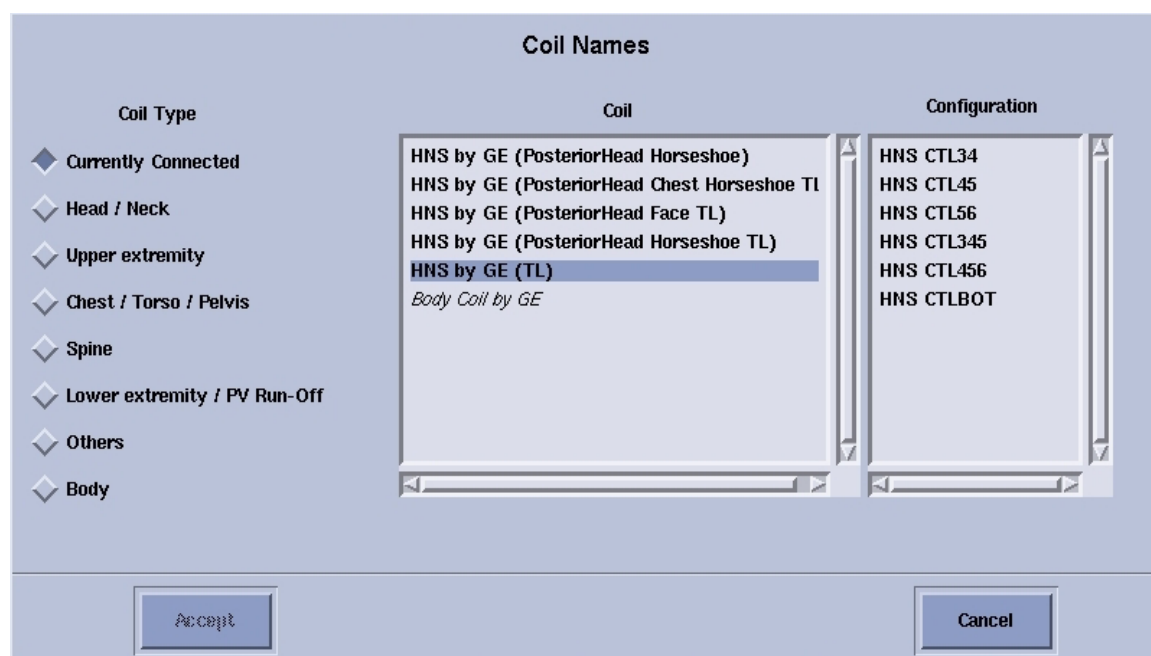
Workflow

To best utilize the HNS Coil, perform the following steps:

1. Set up the coil components necessary to scan the region of interest.

NOTE: Ensure that all the components on the table are plugged in before scanning.

2. Register the patient then select Coil Type “Currently Connected.”
3. Select the best configuration for the region of interest (keep in mind the maximum FoVs for the selected configuration).



NOTE: Some configurations are only compatible when certain coil components are plugged into the system. If the configuration you want is not listed, it may be necessary to add or remove coil components before your configuration is selectable. (Refer to Table 3-1 for a list of configurations and compatible components.)

4. Position the area of interest over the elements in the selected configuration.
5. Landmark to the middle of the element groups used in the selected configuration, then Advance to Scan.

NOTE: Precise patient positioning and proper FoV selection will yield optimum results. **Improper use of the coil is the major cause of image artifacts.** To diminish artifacts, it is important to place the body part to be scanned over the elements in the selected configuration then landmark to those elements. Refer to the section "Coil Mode Configuration" (page 3-14) for guidance.

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Chapter 4

Coil Maintenance

Introduction - Coil Maintenance

To maintain the highest possible image quality, and to protect your patients and yourself from potential injury or illness, it is important to keep the coil clean and well maintained. In this chapter, we discuss basic methods for keeping the coil safe and sanitary. We also provide you with a list of accessories that will enhance the utility of your coil, and you will find contact information in this section should you have questions or comments you would like to pass along.

Quality Assurance

Coil Quality Assurance tests must be run by a GE Service Representative or a third party service provider. To have a Quality Assurance test run on a coil, please call your GE Service Representative or your third party service provider.

If you do not have a GE Service Representative or third party service provider, use the Class A Service CD that was shipped with your MR System. This CD contains information on phantom setup and the usage of the MCQA tool. Please contact GE Healthcare at 800-582-2145 with any questions or concerns.

Coil Cleaning and Storage

Environmental Requirements

When the coil is not in transport and is not being used, store the coil in an area that has environmental conditions that fall within the specifications given under “Operating Conditions” in Table 4-1.

Table 4-1 Environmental Requirements

	Transport/storage	Operating conditions
Barometric pressure	860 hPa to 1060 hPa	
Humidity	20% to 95% non-condensing	30% to 75% non-condensing
Temperature	-15°C to 55°C (5°F to 131°F)	15°C to 35°C (59°F to 95°F)

To store the coil, a storage space that exceeds the dimensions of the coil is required (51 inches wide x 35 inches deep x 18 inches high) (130 cm wide x 89 cm deep x 46 cm high).

The coil should be stored in a horizontal position similar to its orientation on the MRI table. Head Neck Unit cables should be carefully placed through the two cable slots on the posterior and then the anterior should be latched in place. The TLU and NCU cables should be stored in a way that does not expose the cable to kinks or undue stress. The NCU and Anterior Adaptor should be stored flat with no weight placed on top of them.



WARNING:

Detach coil connector from the scanner before attempting to clean the coil. Do not touch connectors with bare fingers. Never press sharp objects against connector surface. Do not reattach connector after cleaning until the coil has dried completely. Electric shock may occur if the coil is attached to the system during cleaning or when it is still wet.



CAUTION:

Do not spray or pour cleaning solution directly onto the coil. Do not submerge the coil in any solution. The coil contains sensitive electronic components that may be damaged by the solution. Under no circumstances should the coil be placed into any type of sterilizer. It should be cleaned only according to the procedure described below.

The HNS Coil and patient comfort pads must be cleaned after each use using following procedure:

- Wipe with a cloth that has been dampened in a solution of 10% bleach and 90% tap water, or 30% isopropyl alcohol and 70% tap water.
- Should your coil need to be returned to GE Healthcare for service, wipe it down with a 10% bleach solution (as described above) to minimize risk of exposure to potentially infectious agents.

The coil should be cleaned only according to the procedure described above.

Accessories

Table 4-2 1.5T and 3.0T HNS Coil Accessories

Part Name	Part Number
Patient comfort pad kit	2418081

Contact Information

GE Healthcare Americas	
USA	800-558-5102
Canada	800-668-0732
GE Healthcare Asia/Australia	
China	86-21-62192228
Taiwan	886-2-2505-7900
Singapore	65-291-8528
Australia	61-2-9975-5501
Japan	81-120-48-2630
Korea	82-31-740-6119
India	91-80-845-2923
GE Healthcare, SCS	33-1-41-19-76-76
GE Website	www.gehealthcare.com
Manufacturer Address	GE Healthcare 1515 Danner Drive Aurora, Ohio 44202-9273 USA
Manufacturer Phone Number	1-330-562-1000

Revision History

Rev.	Date	Author	NPI/ECO #	Primary Reason for Change
1	09/08	Susan Selzer	N/A	First issue
2	01/09	Vishal Kampani	ECO2008333	Update manual with P connector System information
3	N/A	N/A	N/A	Revision 3 was not Released
4	09/09	Dale Seaman	See MyWorkshop	All pgs-updated footer; Title and CE pg-added 3.0T information; Pg 1-ii-changed 'indications for use' to 'intended use' and updated 1.5T uses and added 3T uses, clarified coil description; Pg 1-iii-added "operator manual..." "user must read..." "this manual should..." notes; Pg 1-1-added 3.0T information, added Legacy Bendix to 1.5T coil description in 1st paragraph; Pg 1-3 to 1-6-added labels and definitions to table 1-1; Pg 1-2; added cable and ECG lead verbiage for cable loop warning; Pg 1-3-removed ESD symbol; Pg 1-6-updated format of Coil Classification Table; Pg 1-7 to 1-8-updated part number for positioner; Pg 1-9, 1-10-added 3.0T coil shipping information; Pg 1-7 to 1-11-added box 4 information; Pg 2-1-verbiage in the cable loop warning is now the same as warning on pg 1-2; Pg 2-3-added warnings depicting the location of warning and caution labels on the coil; Pg 3-1-Added 3.0T coil information to Note and Coil Specifications table; Pg 3-3 to 3-10-added new photos and instructions for mirror repair and replacement; Pg 3-11-changed anterior adapter unit to anterior adapter block; Pg 3-12-changed note to caution, also emphasized 1.5T coil and added Medrad and 1.5T and 3T compatibility notes; Pg 3-20-added configuration information and added table 3-2 and changed to 'refer back to table 3-1 for Asset applications'; Pg 3-46-added strap note under warning; Pg 3-49-added 3.0T Legacy connector information, updated NRR rating in Warning to 29dB, revised Bendix connector name to Legacy Bendix connector; Removed QA chapter; Pg 4-1-added QA instruction; Pg 4-2-removed reference to 1.5T coil and added "...coil should be cleaned..." instruction; Cover page and throughout-standardized coil names; throughout-used HNS instead of Head, Neck, and Spine (HNS)